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Techno-economic feasibility and performance analysis of an islanded hybrid renewable energy system with hydrogen storage in Morocco

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ABSTRACT

This study focuses on the conceptual design and viability assessment of a hybrid microgrid system for a settlement in Dakhla city. The system consists of a 600 kW wind turbine, 300 kW diesel generators for backup, a 300 kW fuel cell, and a 500 kW electrolyzer. A simulation model using TRNSYS software was developed to analyze the energy exchange and ensure system reliability. The study incorporates a power dispatch management strategy (PDMS) with load following mode (LFM) and cycle charging mode (CCM) approaches, aiming to enhance the understanding and management of stand-alone renewable energy systems with electrochemical storage. A techno-economic investigation is conducted for a settlement load profile using the hybrid wind/Fuel/DEGS configuration to evaluate its feasibility.

The performance study reveals that the hybrid system achieves a significant renewable fraction of 44.57 %. From an economic perspective, the proposed hybrid system proves to be financially viable. The analysis shows an NPV_{Cost} of \$2,650,843 and $NPV_{Production}$ of 37,819,172 kWh, resulting in a levelized energy cost of \$0.0701/kWh. Additionally, the system demonstrates the potential to reduce CO_2 emissions by 70,529.80 tons over its 20-year lifespan, with a carbon reduction cost of approximately \$37.5847 per t/CO_2e.

These findings highlight the substantial cost savings of approximately \$2,656,754.51, equivalent to the creation of 5256.16 acres or 2125.7 ha of new green space and plants. The combination of economic viability and environmental benefits positions the proposed hybrid energy system as a promising alternative for meeting long-term energy needs while mitigating environmental impact. The study underscores the potential of the system to provide cost-effective electricity to remote communities, thereby contributing to addressing the energy crisis and fostering a sustainable future. The proposed hybrid energy system has significant potential to provide electricity to remote communities at low production costs, as well as to contribute to preventing, controlling, and overcoming the energy crisis for a sustainable future.

1. Introduction

1.1. Background

The demand for electrical energy has been increasing rapidly over the past decade, raising concerns about the negative impact on the environment and the depletion of fossil fuels. As a result, there is a growing interest in renewable energy sources, which are free, abundant, sustainable, and environment-friendly [1]. Hybrid systems (HS), which integrate renewable energy sources and energy storage devices, have emerged as a viable solution for reducing greenhouse gas emissions [2]. These systems can be integrated into microgrids, either as stand-alone systems or connected to the grid. However, energy storage remains a major challenge for hybrid systems.

Numerous studies have investigated the techno-economic feasibility of integrating hybrid energy storage systems. For instance, El Hassani et al. [3,4] conducted a techno-economic evaluation of CSP and PV

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Nomenclature

Latin symbols

A_r	Rotor swept area (m^2)
b_{Tafel}	Tafel slope
B	Lapse rate of temperature (6.5 K per km of altitude)
CO_2	Carbon dioxide
DEGS	Diesel engine generator system
E_g	Annual energy generated by the wind turbine in kilowatt-hours (kWh)
E_r	Rated power of the wind turbine in kilowatts (kW)
E_{aux}	Auxiliary to the renewable energy source
E_{load}	Total amount of electricity serviced (kWh/yr)
F	Faraday constant (F = 96.485C/mol)
HHV_{H_2}	Hydrogen fuel's calorific value (MJ/kg)
$\dot{V}_{Diesel}$	Amount of fuel consumed (m^3/s)
LHV_{diesel}	Fuel's lower heating value (J/kg)
$n_{m,ser}$	Number of fuel cell modules connected in sequence per stack
$n_{c,ser}$	Number of consecutive cells in series
n	Moles of electrons per mole of water
N_{DEGS}	Number of identical diesel units
$\dot{m}_{H_2}$	Hydrogen production's mass flow rate (kg/s)
P_{DEGS}	A single diesel engine electrical energy production
$P_{Total,DEGS}$	Total output power
P_{wind}	Electrical power generated by the turbine
R_{Ohm}	AFC module Ohmic resistance
R	Ideal gas constant
r	Project discount rate
U_{stack}	Voltage across each alkaline fuel cell stack
U_{cell}	The fuel cell voltage
U_0	Open circuit voltage
U_{mod}	Module voltage
T_0	Reference temperature (288 K)
T_z	Temperature at a specific height
X	Normalized power
z	elevation

Greek symbols

C_p	Coefficient of power
η_{el}	Electricity efficiency
η_{ELY}	Electrolyzer efficacy
η_F	Faradays efficacy
n_c	Total number of cells in a sequence
v	Speed of the wind (m/s)
ρ	Density of the air (kg/m^3)
ρ_{diesel}	Density of the fuel (kg/m^3)
ρ_{elev}	Air pressure at a particular elevation
λ	Intercept coefficient

Acronyms symbols

AFC	Alkaline fuel cell
FC	Fuel cell
HRES	Hybrid Renewable Energy System
O&M	Operations and Maintenance
LCOE	Levelized Cost of Energy
NPV_{Cost}	Net present value of total cost
$NPV_{Production}$	Net present value of total production
RF	Renewable Fraction
SOC	State of charge

plants, as well as their combination. The study examined the impact of various design factors such as orientation angles, solar multiple (SM), and thermal energy storage capacity (TES) on yearly energy production, cost, and dispatchability to meet basic load requirements. The results showed that the hybrid plant had a lower LCOE than a CSP alone, and the capacity factor could reach more than 90 % with an SM of more than 3.5, especially for thermal storage of more than 16 h. Bousselamti et al. [5] presented a techno-economic study of hybrid CSP/PV production integrated thermal energy storage (TES). They also investigated the effect of some design parameters on hybrid energy and cost production. The study established that the production of CSP/PV incorporated TES in Morocco was an economically feasible solution in comparison to CSP alone. Similarly, Parrado et al. [6] performed a techno-economic analysis of three 50 MW power plants a PV plant, a CSP plant with a 15-hour TES, and a PV-CSP hybrid plant. They found that the PV-CSP hybrid plant showed better potential to produce an uninterrupted output to cover energy demand at a lower cost.

In addition, several studies have worked on battery storage to compensate for power outages. Barakat et al. [7] conducted a comparative analysis of five different types of energy storage batteries for a PV/battery system connected to the grid in El Dabaa, Egypt. The Hybrid Optimization Model for Electric Renewables (HOMER) software was used to analyze the investment cost, renewable energy fraction, and excess electricity. The study found that the duration of the investment plan has no impact on the proposed system size, renewable energy percentage, CO_2 emissions, or unmet electric load. Samy et al. [8] focused on finding greener and more sustainable technologies to provide electricity in developing communities with inadequate access and unstable grid service. The authors compared the techno-economic performance of photovoltaic energy systems with different battery technologies and suggested a sizing technique for grid-connected photovoltaic systems. The results showed that nickel-iron battery system was the most efficient at high main grid failure frequencies. Likewise, Eteiba et al. [9] conducted a comprehensive techno-economic analysis of an off-grid PV/biomass hybrid system, comparing three different battery technologies. Using various optimization techniques such as Flower Pollination, Harmony Search, Artificial Bee Colony, and Firefly algorithms, they found that the Firefly Algorithm outperformed the other techniques by achieving the optimal configuration with the lowest net present cost while maintaining a specified loss of power supply probability and percentage of excess energy. This highlights the significance of selecting the appropriate optimization technique to enhance the technical and economic feasibility of renewable energy systems. Other studies have proposed innovative electro-thermal energy storage (ETES) technology that can utilize various thermal storage materials, such as thermal oil, molten salt, and sand, at high capacities and improved efficiencies [10]. They found that the use of sand as a thermal storage material enabled the storage system's efficiency to reach 85 %, and the cost is projected to be up to six times lower than that of current Lithium-ion batteries. Jiao et al [11] proposed a consequential life cycle assessment approach to evaluate and forecast the life cycle greenhouse gas emissions from hybrid energy storage systems (HESSs) in 100 % renewable power systems. The study was based on the power system of Sweden and discusses different HESS combinations, which include energy storage (ES) technologies such as pumped hydro ES, hydrogen ES, lithium-ion (Li-ion) batteries, lead-acid (PbA) batteries, vanadium redox (VR) batteries, supercapacitors (SCs), and flywheels. The results show that the Pumped hydro+Li-ion+Flywheel combination exhibits the least life cycle GHG emissions among the HESSs compared in this study.

In addition, several studies have emphasized the importance of hydrogen storage in achieving a 100 % renewable energy system. Marocco et al. [12] investigated the role of hydrogen storage in combination with batteries in an energy system. The study uses a revised OSeMOSYS framework with interconnected clustered representative days and a photovoltaic and floating offshore wind turbine island energy system as a case study. The results show that hydrogen storage is crucial in achieving cost-effective energy system configurations, as it avoids costly oversizing of wind turbines and batteries. The cost of an only-battery system is 155 % higher than that of a hydrogen-based system. Lokar et al. [13] evaluated the annual power generation of a photovoltaic system and a storage battery pilot system applied to a house located in Slovenia. The results showed that the implementation of hydrogen fuel cells into the hybrid system would allow for reaching total self-sufficiency. Similarly, Mert et al. [14] investigated the effect of shading on a standalone PV-Hydrogen system executed in Adana (Turkey) at ATU University. The focus of the study was on performing electrolysis with varying energy supplied to the cell. Hassan et al. [15] conducted a study on an off-grid photovoltaic energy system for hydrogen production. The study analyzed the optimal size of a proton-exchange membrane water electrolyzer fed by a 12 kW photovoltaic array in Baghdad, Iraq. The simulation process considered the project lifespan from 2021 to 2035 and was conducted using MATLAB. The study found that the optimal electrolyzer capacity is 8 kW, producing 37.5 kg/year/kWp of hydrogen for USD 3.23/kg. The results suggest that potentially cost-competitive alternatives exist for systems with market combinations resembling renewable hydrogen wholesale. Saleem et al. [16] proposed a solar hybrid system, consisting of a photovoltaic panel system, fuel cell, hydrogen storage, and a monitor, carried out under different climatic conditions using TRNSYS software. They found that in the cold environment (Fargo), the hybrid system had an efficacy of 7.8 % due to the lack of proper sunshine. However, an efficacy of 11.8 % was achieved in the hot climate (Lahore).

Ghenai et al. [17] utilized TRNSYS simulation to develop a hybrid off-grid-based renewable energy system that supplies the energy load of 150 homes in a desert region (Sharjah). The analysis showed that the system meets the daily life demands of the community, and the techno-economic assessment revealed an adequate level of Renewable Energy System (RES) penetration, with a fraction of 40.2 %, a Levelized Cost of Energy (LCOE) of $145/MWh, and no emissions of CO_2 during the entire electricity production process. Al Ghaithi et al. [18] analyzed the economic viability of off-grid solar PV systems in Masirah Island, Oman. The study showed that a hybrid energy system consisting of wind, photovoltaic, and diesel generators was the most feasible option and improved the voltage profile at the connection point. Qolipour et al. [19] conducted a techno-economic assessment of hydrogen generation using a PV-wind hybrid power plant in the Hendijan environment (southwestern Iran) utilizing Homer software. The hybrid system's yearly electrical energy production was 3,153,762 kWh, with a hydrogen production of 31,680 kg using a system configuration consisting of a GE 1.5sl wind turbine model, a 4 kW photovoltaic system, and a 100-kg hydrogen tank. Hassani et al. [20] proposed an energy management strategy for a photovoltaic system with storage batteries and a fuel cell system. The system was designed to supply a house with a consumption of 2808 kWh/day in Bejaia, Algeria, where the solar potential is 1.2 kWh/m^2/day. A fuzzy logic controller was used to optimize the photovoltaic power, and energy supervision was applied to manage the different sources. Tests were carried out under different meteorological conditions, and the results obtained by simulation were analyzed. Finally, an economic study was conducted using Homer Pro software to highlight the feasibility of the system. Loisel et al. [21] optimized the size of a hybrid wind-hydrogen plant to offer an economic study of hydrogen production with H_2 costs ranging from 4 to €13/kg. Likewise, a techno-economic analysis was conducted in Izmir-Cesme, Turkey, for a hydrogen refueling station powered by wind-PV using HOMER software [22]. The cost of hydrogen generation was estimated to be between $7.5 and $7.9 per kilogram, demonstrating that a solar and wind refueling station for the city would be feasible. A similar investigation was conducted by Shaahid et al. [23] performed a similar investigation for a remote community with an annual electrical energy consumption of 15.943 MWh, where the cost of energy production (COE) of the PV-diesel hybrid system was found to be $0.118/kWh. Similarly, Silva et al. [24] examined the usage of an electrolyzer and a hydrogen tank to the use of batteries as a solar system's energy storage option. According to the findings, batteries are still the best solution for energy storage. Rezaei et al. [25] assessed the wind-hydrogen energy conversion system under different weather conditions of Iran (13 cities in Iran's Fars region). To find the best location for wind-hydrogen energy conversion, the Data Envelopment Analysis (DEA) approach was used. To assess the magnitudes inefficiencies of a hybrid system consisting of a wind turbine and an electrolyzer, Ahmad F. Tazy et al [26] and Charafeddine Mokhtara et al [27] worked on the electrification of residential buildings using hybrid renewable energy systems (HRES) using real data and modeling techniques to evaluate the technical and economic feasibility of HRES configurations and considering factors such as renewable potential, building energy efficiency, and climate diversity. In a similar investigation, Rehman et al. [28] investigated the feasibility of a hybrid wind-PV-diesel power system suitable for a village in Saudi Arabia and found that the most feasible system had an energy cost of 0.212 US$/kWh and consisted of three 600 kW wind turbines, 1000 kW PV panels, and four 1120 kW diesel generators, with a 35 % renewable energy penetration. Dawood et al. [29] evaluated the feasibility of a hydrogen energy storage system for a microgrid hybrid solar PV-battery-hydrogen, while Oueslati [30] simulated a wind-PV-fuel cell system for the Tunisian climate, which included diesel engines as a backup. The system achieved a reduced unmet and excess energy, with an acceptable renewable fraction of 35.52 % and lowered levelized cost of energy (LCOE) of $0.0492/kWh. Samy et al. [31] evaluated standalone hybrid power generation systems using solar, wind, and biomass energy for rural electrification in Saudi Arabia and found that a remote PV/Biomass system was the optimal option, with a TNPC of $138,521.40 and an LCOE of $0.099/kWh. Another study by Samy et al. [32] proposed three combinations of hybrid systems for a small-scale countryside area in Egypt, using optimization algorithms to achieve optimal performance and cost. The PV/Wind/Fuel Cell system with an electrolyzer for hydrogen production demonstrated excellent performance and an economically viable LCOE of $0.47/kWh. Aykut et al. [33] focused on the techno-economics of off-grid wind, solar, biomass gasifier, and fuel cell systems for energy generation and storage. The study proposes a rule-based energy management scheme and an optimization algorithm (Hybrid Firefly Genetic Algorithm) to minimize the annual cost system and meet energy demand reliably. The decision variables for the optimization process are PV panel power, WT power, and the number of H_2 tanks. The results showed that the proposed algorithm has excellent convergence capacity and provides high-quality outputs. Hajiaghasi et al [34] conducted a comprehensive review of the state-of-the-art hybrid energy storage systems (HESS) for microgrid (MG) applications, proposing future trends in HESS for MGs, such as multi-objective design methods, the use of impedance source converters, and investigating the effect of hierarchical control on HESS operation and stability. Jani et al. [35] proposed a two-level energy management strategy to optimize the operation costs of islanded networked microgrids and manage uncertainties. The proposed hybrid energy scheduling considers surplus and shortage powers to determine the optimal transactive energy among microgrids and an autonomous operation scheduling is performed at the lower-level of the optimization framework to minimize the total operation cost of each microgrid. The global optimization is performed at the upper-level to minimize the total operating costs of the networked microgrids. The simulation results showed that the proposed model reduces the operating cost of the system by $174.75.

1.2. Research gap and problem statement

The growing interest in the field of hybrid renewable energy systems (HRES) has seen significant growth in recent years, as evidenced by the abundant literature on numerous algorithms and optimization approaches [36–39,51]. However, despite this progress, the practical implementation and effective operation of HRES remain a challenge. One reason for this is the limited utilization of effective energy management strategies and control systems, which can result in suboptimal performance and hinder the wider adoption of renewable energy systems. Thus, there is a pressing need for further research and development to address these challenges and enhance the implementation of HRES in practical applications.

To fill these gaps and advance the field, this investigation seeks to design and evaluate the performance of a stand-alone HRES in the Saharan region of Dakhla, Morocco, with a focus on meeting specific requirements while minimizing energy losses through the implementation of a comprehensive management and control strategy. The study introduces a power dispatch management strategy (PDMS) that combines load following mode (LFM) and cycle charging mode (CCM) approaches. This innovative strategy is designed to enhance the understanding and management of stand-alone renewable energy systems with electrochemical storage. The proposed HRES includes a wind turbine, a fuel cell, and a diesel generator as a backup. An indirect method will be used to couple the wind turbine to an alkaline electrolyzer to produce cleaner hydrogen.

The investigation focuses on designing and evaluating a Hybrid Renewable Energy System (HRES) for an off-grid residential settlement in Dakhla, Morocco. The system combines wind turbine, fuel cell, and diesel engine technologies to meet the power demand of the settlement. A techno-economic study is conducted to propose an optimized system that improves performance, ensuring the provision of the required amount of energy at the most cost-effective design, increases renewable energy penetration, reduces the Levelized Cost of Energy (LCOE), and minimizes carbon dioxide emissions from both an economic and environmental perspective. Furthermore, this investigation aims to validate and build upon existing research in this field. Through the collection of empirical data, the study will serve to inform renewable energy policy in the region, as well as contribute to the development of sustainable energy systems globally.

The investigation is structured into several sections. The first section provides a general overview of the hybrid renewable energy system, including a detailed description of its components, types, and unit sizing modeled through TRNSYS. The second section focuses on the design of optimal control approaches, namely the Load Following and Cycle Charging Modes. The third section analyzes the results and presents extensive discussions. Finally, the fourth and final section provides the key takeaways and main conclusions derived from this investigation.

2. General overview of the studied system

A typical wind hydrogen system-based standalone composed of WECS-electrolyzer-fuel cell-DEGS was investigated in this study to examine its feasibility to satisfy the required electrical load of an off-grid population in a residential neighborhood situated in the Saharan region of Morocco. The design considered comprises a compressor permitting to distribute of the hydrogen generated by the electrolyzer to the hydrogen store at high pressure, and an inverter to convert direct current to alternating current or vice versa alternating current to direct current. Furthermore, an alkaline electrolyzer was chosen for its outstanding operating conditions. The analysis was carried out in Dakhla city, the choice of the site was based in part on the large wind potential available in this site, and the presence of water sources, which makes it easier to supply the amount of water needed to feed the electrolyzer. However, the standard TRNSYS components that were employed to model the system wind-hydrogen-DEGs, are detailed in the following section.

2.1. Overview of program

The complete investigation model of the wind-fuel cell- diesel system-based mini-grid performance for a standalone application is run using TRNSYS software [40]. It is a comprehensive and versatile simulation platform for dynamic systems simulation. In the context of international collaborations, TRNSYS is today the world reference in the field of dynamic simulation of buildings and systems. It allows you to model the system's energy dynamics. In an HEQ (High Environmental Quality) approach, this type of tool becomes an essential element to validate the energy concept as well as to develop and experiment with innovative approaches (management strategies, renewable energies, architectural renewable energies, architectural variants, etc.). It consists of various components, available in a standard library, which allows for a different simulation problem in completely different domains, from the simplest to the most complex system, including innovative renewable systems technology. However, the program seems to be well suited to study the standalone hybrid wind-fuel cell-diesel power system. Fig. 1 shows a schematic representation of the hybrid system used in this investigation.

2.2. Detail of system components, types, and unit sizing

The investigation's goal is to design and develop a model that intends to demonstrate the implementation of a standalone hybrid power system that can operate without any external power input used in a remote location where transporting electricity is either very difficult or expensive. To study the dynamic behavior of the whole system, a simulation program using the TRNSYS software has been realized giving an overview of the performance behavior of each element of the HRES. The standard TRNSYS components used to create the model are listed below in detail:

2.2.1. WECS

In the current work, type 90 was chosen to simulate the wind turbine (WECS). The input parameters of the model are adjusted to suit the site under study (Dakhla) which is 7 m above sea level. The main characteristics of the 600 kW wind turbine and more details contained in the external file of type 90 are illustrated in Table 1. Fig. 2 depicts the graph of a power curve for a typical Enercon E40/600 kW wind turbine, where the power curve of a wind turbine displays the amount of electrical power generated by the turbine at various wind speeds. Theoretically, the equation for expressing the wind turbine's power output, P_{wind}, is given by the formula below [30]:

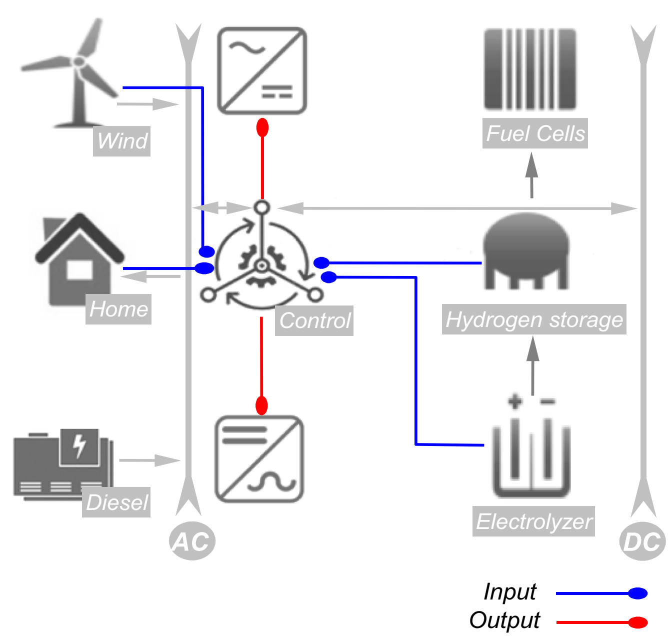

Fig. 1. A schematic workflow of a hybrid wind-fuel cell-diesel system.

Table 1
The main characteristic of the concerned WT.

Item description	Value	Unit
WECS type	Enercon E40 600/46	–
Rated power	605	kW
Speed power	13.50	m/s
Rotor height	46	m
Rotor diameter	43.7	m
Sensor height	46	m
Site shear exponent	0.14	–
Turb int	0.10	–

$$P_{wind} = \rho C_p A_r v^3 \tag{1}$$

where:

ρ: The density of the air (kg/m^3)
v: The speed of the wind (m/s)
A_r: The rotor swept area (m^2)
C_p: The coefficient of power

The density of air varies with temperature and pressure as follows [30]:

$$\rho_{elev} = \frac{P_{elev}}{RT} \tag{2}$$

The temperature decreases linearly with elevation for the heights where wind energy facilities will be implemented.

The "lapse rate" of temperature is given by [30]:

$$T_z \approx T_0 - Bz \tag{3}$$

where $T_0 = 288K$ and $B = 6.5K$ per km of altitude.

In addition, the relationship between elevation and wind speed variation is expressed as follows [30]:

$$\frac{v_1}{v_2} = \left(\frac{z_1}{z_2}\right)^\alpha \tag{4}$$

However, in this research work, the value site shear exponent is 0.14, which is regarded as optimal boundary layer conditions for the particular context of the Dakhla city region.

2.2.2. Data files

The data used in the present study are of Dakhla (23.6848° N, 15.9579° W) over ten years 2010–2020 recording, Dakhla is one of the windiest sites in the world, it has a high wind speed offering good prospects for the installation of wind turbines. In addition, Morocco is currently working on the installation of a 900 MW wind power plant in the Western Saharan city of Dakhla, which will be one of the largest in the world. In this spirit, the study comes to examine the feasibility of producing hydrogen from wind hybrid systems in the city. However, the inputs to the WECS come from both the weather data reader/processor (Type15) and the data reader (Type9). Type 15 is used for reading meteorological data (temperature, atmospheric pressure, and solar radiation for an average year) in the EPW format from Meteonorm/Energy Weather files, While Type 9 is used to generate wind data in real-time from NASA's Surface and Solar Energy webpage, using the exact geographic coordinates of the desired site which are presumed to be measured as hourly average values.

To highlight the overall pattern of climatological wind mapping and its behavior. The figure displays the wind map of Morocco, it is demonstrated that a significant spatial variability of wind speed occurs along the Moroccan Atlantic area (Fig. 3), with the highest values observed in the south. The selected site (Dakhla), located on the southern coast of Morocco, presents favorable wind speeds, especially when analyzing the hourly wind speed intensity.

Furthermore, the monthly average wind speed of data gathered at a height of 10 m in the city of Dakhla is shown in Fig. 4. The wind speed profiles show a large amount of variation. The greatest monthly average wind speed was recorded in July (7.8 m/s). The extracted wind data confirms that the city has a good wind potential. Therefore, it represents an encouraging site for the use of wind energy to produce electricity and will therefore cover the profile load discussed.

2.2.3. Load

The load was defined using two Type 14 forcing functions, one for the monthly load and the other for the daily load. In an equation block named "load," the two profiles are mixed. Note that, the suggested hybrid systems sizes are depending on the community settlement, composed of 250 houses, five small supermarkets, and 200 LED streetlights (Table 2). A single house is expected to consume an average power demand of 1.25 kW (30kWh/day) meaning an average daily power of 312.5 kW for the 250 houses. Similarly, a 7 kW of power load is demanded by the five small supermarkets (1400 W for a single one), while a 500 W is required by the 200 LED streetlights installed within the settlement. Accordingly, an average power load of 320 kW with 639 kW of peak demand is supposed for the micro-grid settlement which means a daily electrical consumption of 7680 kWh/day with a 5 % of percentage day-to-day random variability due to fluctuation of the load demand during the year. Fig. 5 displays the daily profile for a typical day in January and the monthly load of the settlement during the year. As can be seen, during the cold months, the demand increases, which is a result of the additional energy demand required in this specific season, especially for area heating in the residential and commercial sectors.

2.2.4. Mini-grid controller

There is a multitude of ways to control a mini-grid, just as there are diversities of mini-grids. A type 105 model is used to control the mini-grid, which has a bunch of assumptions. This type of component

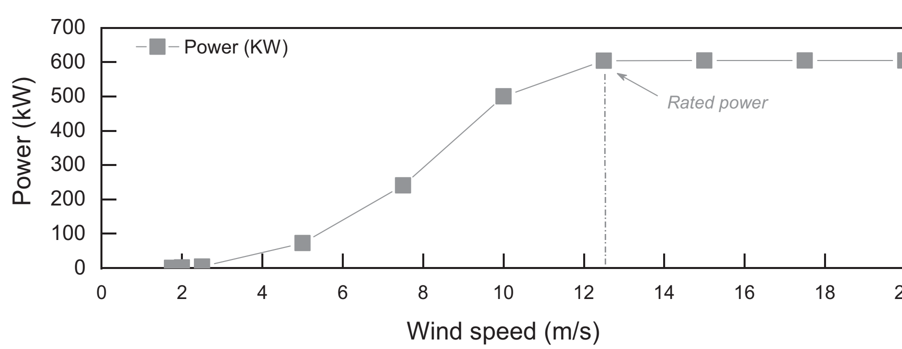

Fig. 2. ENERCON E40/600 model's turbine output variation with wind speed.
Note that the wind turbine used takes into consideration the variation of air density and the increase of wind speed with site altitude. Due to the fact that at a particular constant altitude, the air density tends to fluctuate with temperature and pressure, based on the ideal gas law.

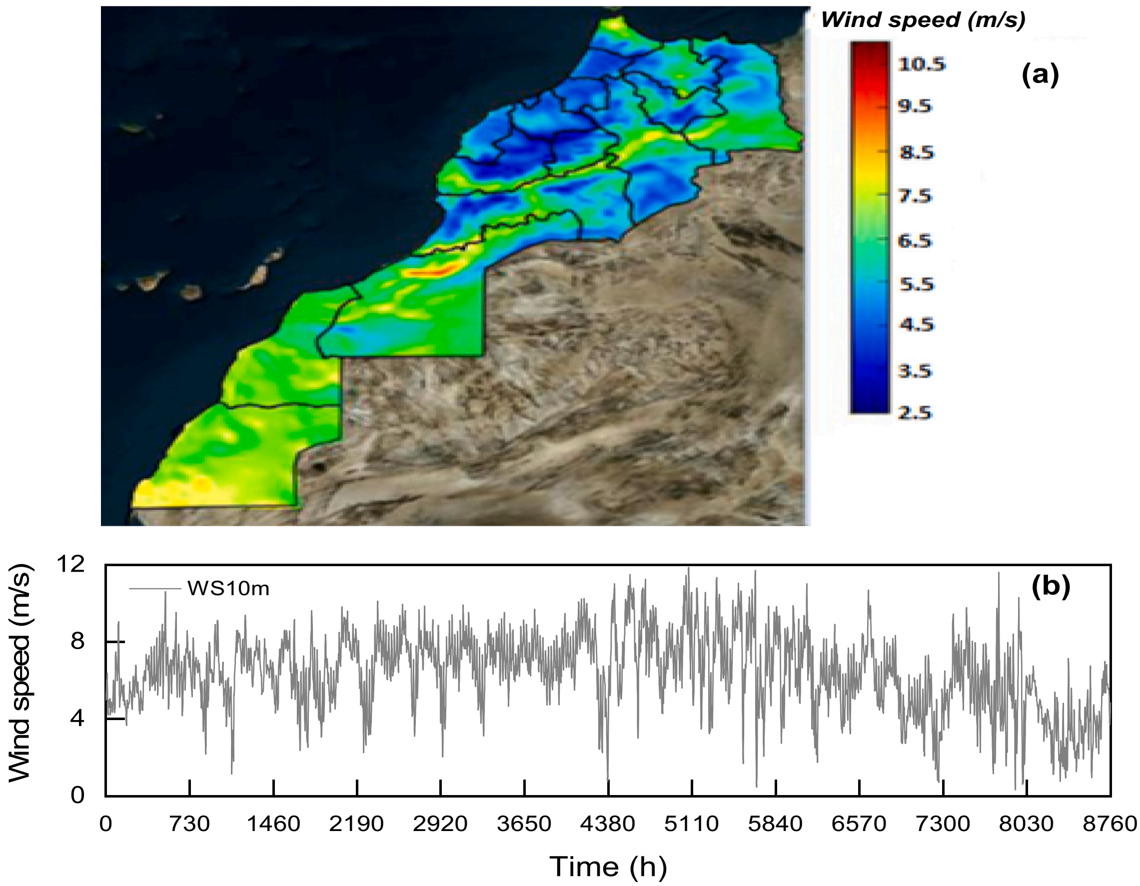

Fig. 3. Wind speed data, (a) spatial distribution in Morocco [41], (b) hourly data in Dakhla city.

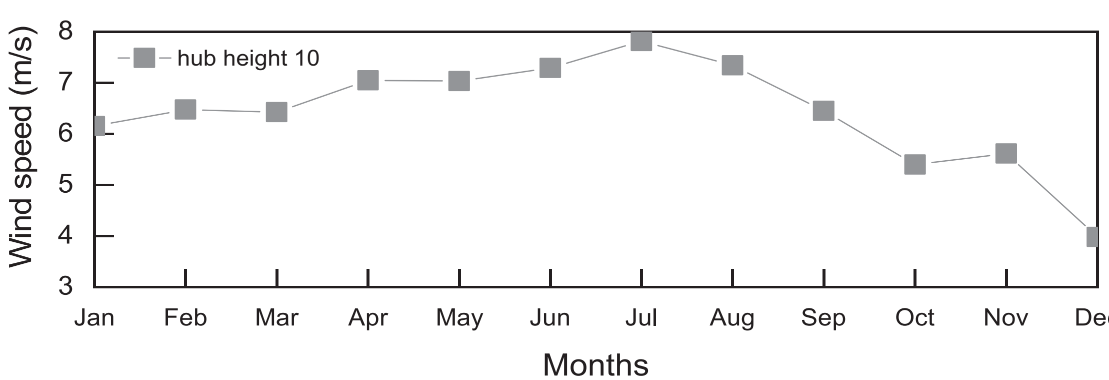

Fig. 4. Wind speed in Dakhla city at a height of 10 m, on a monthly basis.

involves some hierarchization of what should be done and how the load satisfaction should be arranged and controlled between renewable sources, backup sources, the electrolyzer, and hydrogen storage while taking into consideration the demand load. Table 3 provides the core parameter specifications of the mini-grid controller, which plays a crucial role in controlling and managing the operation of the hybrid renewable energy system. These parameters define the operational limits and settings for various components, such as the DEGS (Diesel Engine Generator Sets), fuel cell, and electrolyzer, based on specific conditions and requirements.

The parameters listed in Table 3 offer valuable insights into the operation and control of the mini-grid system, enabling effective management of power flow and load satisfaction. The maximum allowable number of DEGS units, set at 5, indicates the system's capacity to simultaneously operate multiple Diesel Engine Generator Sets. This allows for greater flexibility and reliability in meeting the energy demands of the mini-grid. Each DEGS unit is rated at 300 kW, highlighting its maximum power output capability, which is crucial for determining the system's overall power capacity.

The fuel cell idling power, specified as 60 kW, represents the power consumed by the fuel cell when it is not actively generating electricity. This parameter is significant in understanding the baseline power consumption of the fuel cell and helps assess its efficiency during periods of low demand. With a rated power capacity of 300 kW, the fuel cell demonstrates its ability to deliver substantial power to the mini-grid when needed, ensuring a stable and continuous energy supply.

The electrolyzer idling power, denoted as 120 kW, indicates the power consumed by the electrolyzer during periods of inactivity when

Table 2
Electrical appliances considered in the typical micro-grid community describing the power load demand.

Load type	Number	Appliances	Average electricity consumption (W)
Households	250	Refrigerators	200 × 250
		Air conditioners	500 × 250
		Computers	80 × 250
		Water heater	240 × 250
		4 incandescent lamps (4 * 50 W)	200 × 250
		Television	30 × 250
		Total	312,500
Small supermarkets	5	Big refrigerators	500 × 5
		Small refrigerators	200 × 5
		Air conditioners	500 × 5
		Lightning (4 incandescent lamps (4 * 50 W)	200 × 5
		Total	7000
Lightning	200	LED lights	500

hydrogen production is not required. Understanding this idle power consumption is crucial for optimizing the system's energy usage and minimizing wastage. The rated power capacity of the electrolyzer, specified as 500 kW, reflects its ability to produce hydrogen efficiently and meet the demand for stored energy during periods of low renewable energy availability.

The upper and lower State of Charge (SOC) limits for hydrogen storage provide guidelines for maintaining an optimal range of hydrogen levels in the system. The specified upper SOC limit of 90 % when the electrolyzer is idle ensures that hydrogen production ceases once the storage capacity reaches this threshold. Similarly, the lower SOC limit of 80 % indicates when the electrolyzer is activated to resume hydrogen production. These SOC limits enable efficient management of hydrogen storage, ensuring an adequate supply for the fuel cell during periods of low renewable energy generation.

Additionally, the upper SOC limit of 20 % when the fuel cell is idle ensures that hydrogen consumption stops once the storage capacity falls below this threshold. The lower SOC limit of 10 % indicates when the fuel cell is activated to resume power generation from stored hydrogen. These limits help maintain a balance between hydrogen production and consumption, optimizing the usage of available resources and ensuring an uninterrupted power supply.

By considering these core parameter specifications, the mini-grid controller can effectively prioritize and regulate the flow of energy between renewable sources, backup sources, the electrolyzer, and hydrogen storage, while fulfilling the demand load. This level of control allows for efficient utilization of available resources, maximizes the system's renewable energy integration, and ensures reliable power supply to the mini-grid.

2.2.5. Fuel cell and power converter

The TRNSYS type 173 fuel cell model utilized in this work is a simple mathematical model for an alkaline fuel cell (AFC) that does transform hydrogen chemical energy into direct electrical current in such a manner that hydrogen and oxygen undergo a reaction without causing combustion in an electrochemical reaction (reverse of electrolysis). The fuel cell reactions are represented by the equations below, which depict the anodic, cathodic, and global processes, respectively.

$$\text{Anode}: H_2(g) \rightarrow 2H^+(aq) + 2e^- \tag{5}$$

$$\text{Cathode}: 2H^+(aq) + 2e^- + {}^1\!/\!{}_2 O_2(g) \rightarrow H_2O(l) \tag{6}$$

$$\text{Overall}: H_2(g) + {}^1\!/\!{}_2 O_2(g) \rightarrow H_2O(l) \tag{7}$$

Table 3
Core parameter specification of the mini-grid controller.

Item description	Value	Unit
The maximum allowable number of DEGS in operation	5	–
Rated power of each DEGS	300	kW
Fuel cell idling power	60	kW
Rated fuel cell power	300	kW
Electrolyzer idling power	120	kW
Rated electrolyzer power	500	kW
Upper SOC limit (H_2-storage), ELY switched OFF (idle)	90	%
Lower SOC limit (H_2-storage), i.e., ELY switched ON after having been idling	80	%
Upper SOC limit (H_2-storage), i.e., FC switched ON after having been idling	20	%
Lower SOC limit (H_2-storage), i.e., FC switched OFF (idle)	10	%

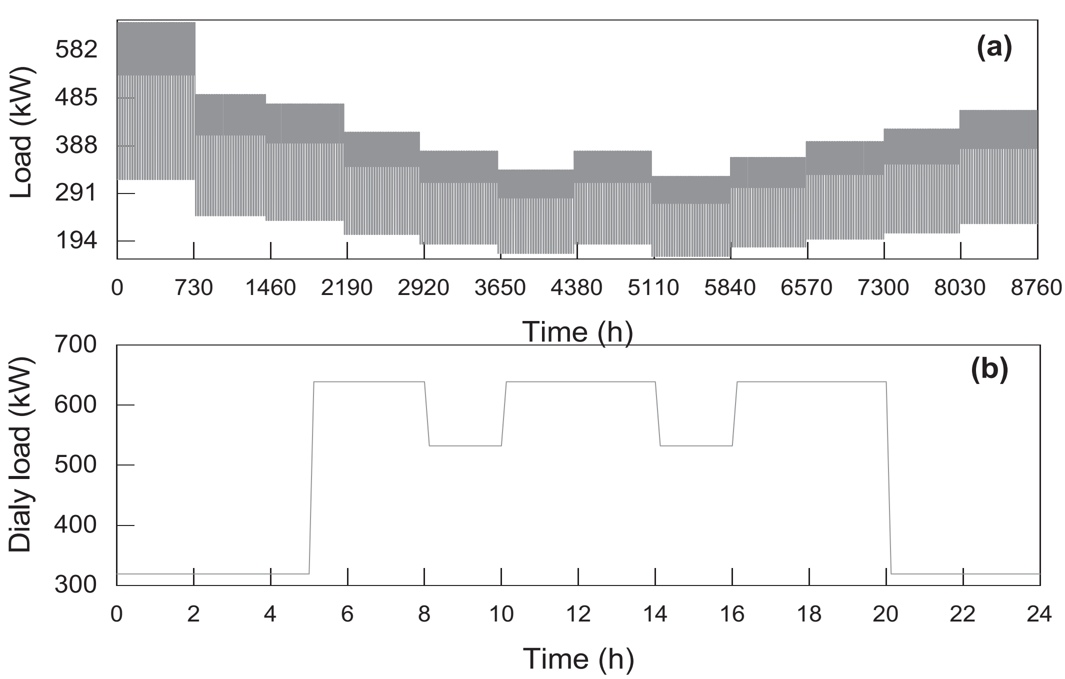

Fig. 5. (a) Monthly load during one year and (b) daily load for the particular community settlement in Dakhla city.

The fuel cell voltage U_{cell} is calculated by dividing the module voltage by the number of consecutive cells as expressed in the eq. [30]:

$$U_{cell} = \frac{U_{mod}}{n_{c,ser}} \tag{8}$$

The voltage across each alkaline fuel cell stack U_{stack} can be found as follows [30]:

$$U_{stack} = n_{m,ser} * U_{mod} \tag{9}$$

where:

U_{mod}: Module voltage.
$n_{m,ser}$: Number of fuel cell modules connected in sequence per stack.

$$U_{mod} = U_0 - b_{Tafel} * log(I_{stack}) - R_{Ohm} * I_{stack} \tag{10}$$

U_0: Open circuit voltage.
R_{Ohm}: AFC module Ohmic resistance.
b_{Tafel}: Tafel slope.

2.2.6. Electrolyzer and power converter

In this study, alkaline water electrolysis (AWE) is described as a device that utilizes electrical power to separate water into oxygen and hydrogen to generate green (clean) hydrogen which can then be stored and utilized by the fuel cell later. However, an electrolyzer needs to be kept on standby power for that reason a type 105 controller was used. The type 160 employed in this study was chosen because of its rapid reaction to power fluctuations, as well as its performance and ability to attain acceptable efficiency in both partial and full loads.

The total P_{ELY} used by the alkaline electrolyzer is calculated [30]:

$$P_{ELY} = \frac{\dot{m}_{H_2} HHV_{H_2}}{\eta_{ELY}} \tag{11}$$

where

$\dot{m}_{H_2}$: Hydrogen production's mass flow rate (kg/s)
HHV_{H_2}: Hydrogen fuel's calorific value (MJ/kg)
η_{ELY}: Electrolyte's efficacy.

Furthermore, Faraday's theory states that the rate of hydrogen generation ($\dot{n}_{H_2}$) in an electrolysis cell is proportional to the rate of electron transport between the electrodes. As a result, in an electrolyzer with numerous cells in series, the overall hydrogen generation rate ($\dot{n}_{H_2}$) may be expressed as follows [30]:

$$\dot{n}_{H_2} = \eta_F \frac{n_c I_{ely}}{nF} \tag{12}$$

where:

η_F: Faradays efficacy
n_c: Total number of cells in a sequence
n: Moles of electrons per mole of water ($n = 2$)
F: Faraday constant (F = 96.485C/mol)

2.2.7. Hydrogen storage

One viable option for energy storage is the utilization of hydrogen (H_2) tanks, which offer a reliable means of storing chemical energy over extended periods. Hydrogen is readily produced and its conversion into heat and electrical energy is environmentally clean. In the current study, a compressed gas storage type 167 was engaged to simulate hydrogen storage, which is considered the most mature technology for this purpose. This type of storage allows the hydrogen to be stored at a specific outlet pressure of 200 bar. The hydrogen produced by the electrolyzer is compressed and buffered in a hydrogen tank, where it can be later used for power generation.

To achieve the desired pressure in the hydrogen tank, a combination of a "compressor setpoint" equation with a type 167 storage system is employed. This ensures that the incoming H_2 gas is compressed to the required pressure. However, it's important to note that there is a maximum permitted pressure of 400 bar. If the pressure exceeds this limit, the excess hydrogen is discharged for safety reasons.

Furthermore, it is assumed in the study that the hydrogen tank will be maintained at a temperature of 20 °C. This temperature condition ensures the stability of the stored hydrogen and its energy content over time.

Additionally, the lower limit of the H_2-storage State of Charge (SOC) has been set at 10 %. When the controller determines that the SOC has reached this threshold, it shuts off the fuel cell, and an alternative power source such as a diesel engine is used to supply the electrical load.

Overall, the proposed system utilizes hydrogen storage in a compressed gas tank, allowing for efficient and clean energy storage and subsequent power generation. Safety measures are in place to prevent pressure build-up beyond the designated limit, and the SOC threshold ensures optimal utilization of the stored hydrogen.

2.2.8. DEGS model

Type 120 was chosen as an auxiliary backup power source in its "generic DEGS" form, which will be brought online when the energy generated by wind and hydrogen is insufficient to fulfill the present demand. For the current study, the lower limit of the power generated by the DEGS is set to 0 kW, while the upper limit is set to 360 kW, implying that the generator is allowed to produce any amount of power between 0 kW and 360 kW. Indeed, the number of DEGs that may be in use was limited to between 0 and 5 units. It is worth pointing out that Type 120 can predict the number of identical individuals of DEGS in operation and then calculate the energy and electrical performance and efficiencies.

The total output power $P_{Total,DEGS}$ is calculated by multiplying a single unit's electrical output power P_{DEGS} by the set number of DEGS units, as follows:

$$P_{Total,DEGS} = N_{DEGS} * P_{DEGS} \tag{13}$$

The DEGS model under consideration is based on an empirically determined relation between fuel consumption and normalized electrical output power [30].

$$\dot{V}_{Diesel} = \lambda + \gamma X \tag{14}$$

where X normalized power is defined as [30]:

$$X = \frac{P_{DEGS}}{P_{DEGS,rated}} \tag{15}$$

The electrical efficiency can be analyzed using the eq. [30]:

$$\eta_{el} = \frac{P_{DEGS}}{\rho_{diesel} * \dot{V}_{Diesel} * LHV_{diesel}} \tag{16}$$

where:

η_{el}: efficiency of converting fuel energy into electricity
ρ_{diesel}: density of the fuel (kg/m^3)
$\dot{V}_{Diesel}$: amount of fuel consumed (m^3/s)
LHV_{diesel}: fuel's lower heating value (J/kg)

3. Design of optimal control approaches

3.1. Power Dispatch Management Strategy (PDMS)

Effective energy management is crucial for the long-term sustainability of hybrid systems consisting of multiple renewable energy sources, fuel cell power generators, batteries, and diverse loads such as hydrogen fueling loads. In this study, a comprehensive model of a wind-fuel cell-diesel hybrid system was developed to design the optimal operation of the system and meet the energy demand of a residential neighborhood. This model, known as the Power Dispatch Management Strategy (PDMS), encompasses two derived approaches for simulating the Hybrid Renewable Energy System (HRES), as shown in Fig. 6(b). The first approach is the Load Following Mode (LFM), while the second one is the Cycle Charging Mode (CCM).

In the PDMS approach, the hybrid wind-fuel cell-diesel system is controlled in a way that utilizes wind energy to meet the settlement's load requirements. When the wind-generated energy exceeds the demand, the excess energy is diverted to an electrolyzer, which produces hydrogen. This hydrogen is then stored in a tank for future use. Conversely, when the wind-generated energy is insufficient to meet the demand, a fuel cell stack is activated to generate power by consuming the stored hydrogen. Additionally, a diesel-powered generator serves as a backup power source when the hydrogen tank is low and the available wind energy cannot fulfill the load requirements. The flow chart of the wind-electrolyzer-fuel cell system is illustrated in Fig. 6. To ensure efficient energy utilization, the following considerations were taken into account:

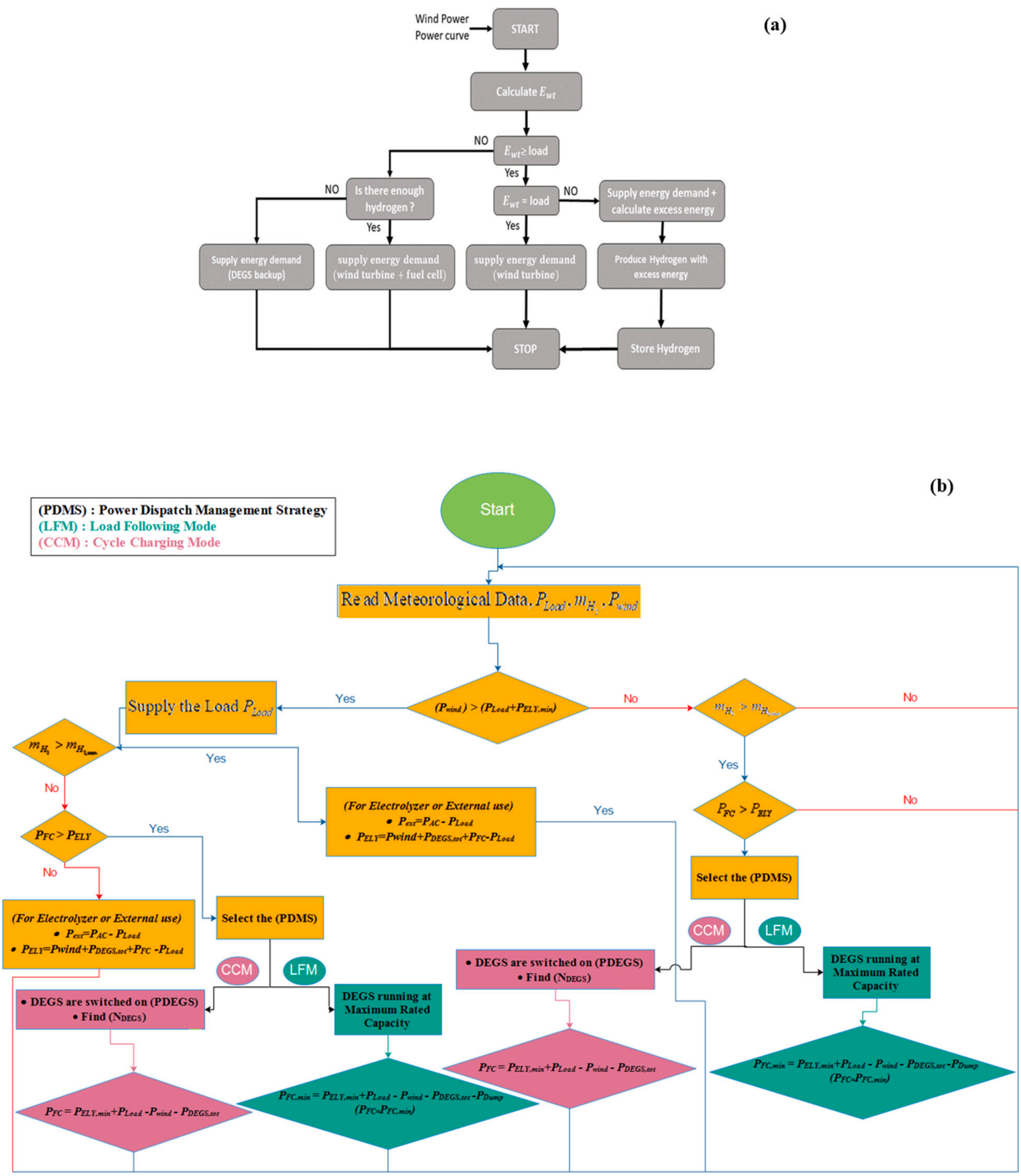

Fig. 6. (a) Flowchart of the Power Dispatch Management Strategy (PDMS) approach used in simulating the wind electrolyzer-fuel cell system and (b) illustration of both dispatch strategies LFM and CCM used in TRNSYS simulation.

- If wind energy exceeds demand, the extra energy will be utilized to produce hydrogen in the electrolyzer.
- If the amount of wind energy available was insufficient to meet demand, a fuel cell was employed to create power from hydrogen stored in the storage tank.
- If wind power is insufficient to satisfy demand and there isn't enough hydrogen in the storage tank, a diesel-powered generator will step in to supply the need.

In addition to the primary approach, secondary strategies are implemented to enhance the accuracy of simulation results. The Load Following Mode (LFM) and the Cycle Charging Mode (CCM) are used simultaneously. In the Load Following Mode, the wind turbine primarily fulfills the energy demand while the Diesel Engine Generator Set (DEGS) system remains inactive when there is no excess power available. When wind energy falls short of the load demand, the fuel cell operates to deliver the required power, and the diesel engines start operating to provide additional power. The Type 105 subroutine, integrated into the TRNSYS system, plays a crucial role in controlling the electrolyzer, fuel cell, and diesel engine generator set, allowing for accurate calculations of their power requirements and determining the number of diesel engines that need to be in operation.

The Cycle Charging Mode (CCM) differs from the Load Following Mode (LFM) in terms of the DEGS system operation. In CCM, when the diesel generator is activated, it operates at its full rated power, and the fuel cell runs at maximum power to meet the load demand. The operation of the electrolyzer and fuel cell is based on an ON/OFF switching mechanism, taking into account the hysteresis effects. The OFF mode indicates that the unit is idling until the pressure level of the hydrogen storage, or the state of charge (SOC), reaches a predefined set point.

Overall, the Power Dispatch Management Strategy (PDMS), along with the secondary strategies of LFM and CCM, enables effective control and coordination of the wind-electrolyzer-fuel cell-diesel system. These strategies ensure efficient utilization of available resources, optimal power generation, and reliable supply to the mini-grid.

4. Assessment and analysis of hybrid energy system performance

Understanding the individual energy potential of each component, as well as their behavior and alignment with community needs, is essential for informed decision-making regarding hybrid energy systems. In this study, TRNSYS (version 18) commercial software was employed to analyze the energy behavior of a wind-fuel cell-diesel hybrid system's components under the meteorological conditions of Dakhla, Morocco (23° 43′ 00″ north, 15° 57′ 00″ west). This allowed for an assessment of the system's feasibility and year-round energy consumption and production on-site. The energy outputs of the main components over the simulation period are summarized as follows.

4.1. WECS simulation result

The hub height of the wind turbine plays a crucial role in analyzing the wind power system in the investigated site. Based on the theoretical model described in the previous section, the wind turbine data collected in Dakhla over a one-year period at a height of 10 m above ground level was modeled with a hub height of 50 m. The simulation results indicate that by increasing the hub height, the wind speeds are higher and more consistent. As seen in Fig. 7(a) at a hub height of 50 m, the monthly average wind speeds exhibit an upward trend, ranging between 9.75 m/s and 4.5 m/s. In contrast, at a hub height of 10 m, the wind speeds show a narrower range, varying from 7.81 m/s to 3.85 m/s. This observation highlights the significance of increasing the hub height to improve the wind resource potential. Furthermore, the monthly evolution of wind speeds in Dakhla demonstrates consistent wind behavior throughout the year. This consistency indicates the presence of reasonable wind conditions in the location, which are conducive to reliable wind turbine output Fig. 7(b). Hence, the simulation results emphasize the importance of considering the hub height when assessing the wind power potential of the investigated site. These findings support the feasibility and desirability of implementing wind systems in the area, taking advantage of the favorable wind conditions observed at the elevated hub height.

4.2. Alkaline electrolyzer simulation results

The alkaline electrolyzer simulation results presented in this study demonstrate the potential advantages and commercial viability of advanced alkaline electrolyzers for stationary applications. Alkaline electrolyzers have been widely used for hydrogen generation through the process of water splitting, making them a well-established technology in the field.

The specific alkaline electrolyzer considered in this work exhibits an electrical power requirement ranging from 115 kW to 430 kW. This power range enables the electrolyzer to produce hydrogen at a rate of 30–100 m^3/h, along with oxygen production at a rate of 15–50 m^3/h, as depicted in Fig. 8. These production rates highlight the electrolyzer's capacity to generate substantial amounts of hydrogen, which can be subsequently utilized for various applications, such as fueling an alkaline fuel cell (AFC).

It should be noted that as a water electrolyzer operates, it generates heat, requiring an auxiliary cooling system of 65 kW to cool the electrolyzer system. Hence, to maintain the efficient operation of the electrolyzer, it is crucial to manage the heat generated during the electrolysis process. In this study, an auxiliary cooling system with a capacity of 65 kW is employed to dissipate the excess heat generated by the electrolyzer. This cooling system ensures that the electrolyzer operates within the desired temperature range, enhancing its overall performance and longevity. The hourly simulated results of the auxiliary cooling power, the heat created by the electrolyzer, the oxygen production rate, the hydrogen production rate, and the total energy drawn by the electrolyzer are shown in Fig. 8.

4.3. AFC simulation results

Replacing traditional hybrid systems with a fuel cell/electrolyzer can significantly reduce air and noise pollution while also exploiting renewable resources more efficiently. In this study, the fuel cell was controlled to create power from hydrogen stored in the storage tank when the wind system fails to supply adequate electricity.

To assess the overall performance of the AFC, a simulation was conducted to understand the behavior of Alkaline fuel cells. The core outputs of AFC are represented in Fig. 9. The results show that the total heating capacity of the FC to run the electrochemical reaction is 40–350 kW per hour. The total power output is 60–300 kW per hour, providing valuable electrical and thermal energy. The amount of hydrogen used by the AFC varies between 0 and 181.82 Nm^3/h. It is worth noting that the heat supplied by the fuel cell can be utilized for hot water or space heating, increasing the efficacy of the fuel cell from approximately 50 % to over 90 % (excluding losses due to parasitic loads and the DC/DC converter).

4.4. DEGS simulation results

The results of the DEGS simulation are presented in Fig. 10, which displays the total waste heat, liquid fuel consumption, and total energy consumption. It is important to note that the DEGS is designed to turn on when wind power is insufficient or when hydrogen storage capacity cannot satisfy the SOC. The 300 kW DEGS, consisting of five units, powers both the baseload and the alkaline electrolyzer for hydrogen synthesis.

The simulation results indicate that the DEGS achieved maximum

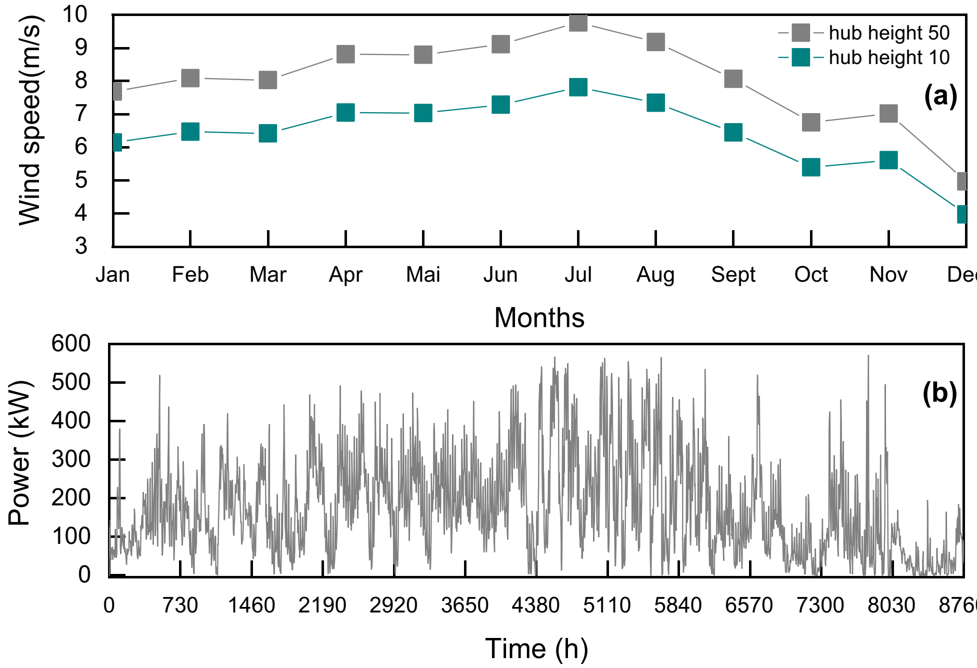

Fig. 7. (a) Monthly average wind speed and, (b) wind power output throughout the year.

efficiency in terms of total power supply, waste heat, and total liquid fuel consumption rate during winter months when wind speeds are low. The reverse is also true. The decision to power the fuel cell stack and the electrolyzer using the diesel generator primarily depends on the future net load, which is the difference between the micro-grid load demand and the output power from the renewable wind source. Predicting the future net load is challenging, so the current net load is used with a combination of both approaches as a proxy for the future net load to determine whether to charge the AFC and electrolyzer from the DEGS.

This strategy tends to use the LFM approach when the net load is high and the primary strategy when the net load is low. During a low net load period, the primary strategy avoids using the generator, while the LFM strategy ensures continuous utilization of the DEGS during high net load periods.

4.5. Total evaluation results of wind/FC/DEGS system

In light of numerous tests and simulations carried out, highlighting the most efficient configuration available for the wind/AFC/DEGS system evaluation. Settlement community charges and wind resources availability are taken into account in the assessment, which is conducted using TRNSYS software. The simulation results include energy production from the wind turbine, diesel generators, and fuel cells, as well as load demand, electrolyzer energy consumption, and energy losses in AC/DC and DC/AC converters, all of which are considered by TRNSYS.

The capacity factors (C_f) that can aid in the prediction of WECS delivery efficiency are expressed as the ratio of the annual energy generated (E_g) by the wind turbine to its rated power (E_r) multiplied by the number of hours in a year (8760):

$$C_f = \frac{E_g}{E_r} \tag{17}$$

The resulting capacity rating of the wind turbine employed in the current study is 600 kW, whereas its average expected energy yield is 264.10642 kW per year. The capacity factor of the supplied WECS Type Enercon E40 600/46, is anticipated to be roughly 44.02 % with an annual energy generation of 2313.50408 MWh.

For the alkaline fuel cell, the average electrical output is reported as 92.729 kW, but the highest recorded electrical production yield was 299.729 kW. The fuel cell has an annualized power production of 812.3989 MWh/year, which is equivalent to an average daily energy production of 2224.6575 kWh/day, assuming an efficiency of 56.6 %.

Concerning the diesel engine model, the system is rated at 1.5 MW (equivalent to 5 DEGS) and generates a total of 1611.4875 MWh per annum. The diesel generator system demonstrates an efficacy of 40.75 %.

In addition, during the continuous 8760-hour service, the AC/DC converter linked to the electrolyzer produced an average energy output of 201.267 kW, with a peak output of 472.579 kW. The input and output energies of the converter were approximately 1831.745 MWh/year and 1763.132 MWh/year, respectively. Meanwhile, the annual energy loss was estimated to be around 68,612.736 kWh, with an average efficiency of 95.91 % per year.

As for the DC/AC converter linked to the fuel cell, an annual input energy of 814.274 MWh/year and an annual output energy of 790.637 MWh/year were obtained, with a peak power of 295.793 kW. The total yearly energy loss of the DC/AC converter was calculated to be 23,637.507 kWh per year. Consequently, an average efficacy of 96.53 % was attained.

According to Fig. 11, the overall generated power for the proposed hybrid renewable energy system is 4714.559 MWh/year, of which 2313.50408 MWh/year comes from the wind generator, which accounts for 49.07 % of the total power generated by the system, and 789.5676 MWh/year comes from the fuel cell, which accounts for 16.75 %. Furthermore, the DEGS provides the anticipated amount of power required for yearly standby, which is 1611.4875 MWh/year, or 34.18 % of the total energy generated by the system.

Furthermore, as shown in Fig. 11, the electrolyzer consumes 1763.05914 MWh every year, accounting for 37.40 % of the overall hybrid system production. More interestingly, the entire system losses due to converter procedures amount represent just 92.2427 MWh per year (1.95 % of the total output).

An important parameter to examine in analyzing the appropriateness of the proposed HRES design is the so-called renewable fraction (RF), which refers to the energy supplied by renewable energy sources to meet the load requirements, which is derived as follows:

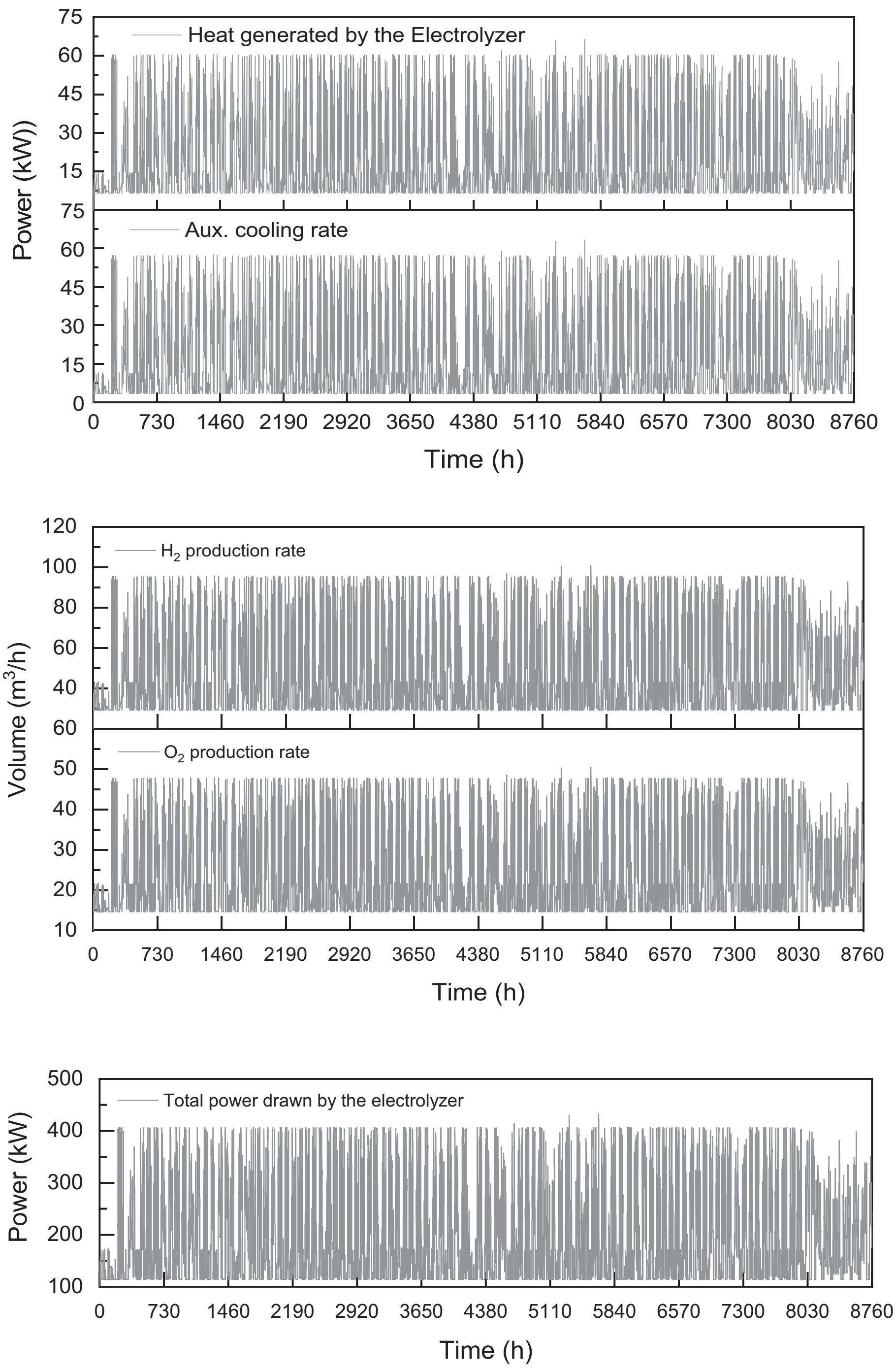

Fig. 8. Core outputs of the electrolyzer simulation result throughout the year.

$$RF = 1 - \frac{E_{aux}}{E_{load}} \tag{18}$$

where E_{aux}: the auxiliary to the renewable energy source, which happens to be the diesel engines (kWh/yr) in the current study.

E_{load}: the total amount of electricity serviced (kWh/yr).

The proposed hybrid renewable energy system (HRES) has demonstrated exceptional performance in meeting the load requirements of the settlement, showcasing a remarkable level of renewable energy penetration with a high renewable fraction (RF) of 44.57 %.

Comparing the findings of other studies, Ghenai et al. [17] achieved an RF of 40.2 % in their off-grid solar PV/fuel cell power system integrated with a solar-based electrolyzer. This indicates that 40.2 % of the energy supplied in their system to meet the load requirements originated from renewable sources. Awan et al. [48] reported an RF of 37.1 % for their PV-diesel-battery system, which was recognized as the most economically efficient hybrid system. This implies that 37.1 % of the energy consumed by their system was derived from renewable sources. In another study by Aziz et al. [49], a PV/diesel/battery hybrid system was investigated, yielding varying RF values depending on the operational strategy employed. The Load Following strategy achieved an RF of 44.7 %, indicating that 44.7 % of the total energy production in that mode originated from renewable sources. The Cycle Charging strategy exhibited an RF of 18.4 %, indicating a lower utilization of renewable

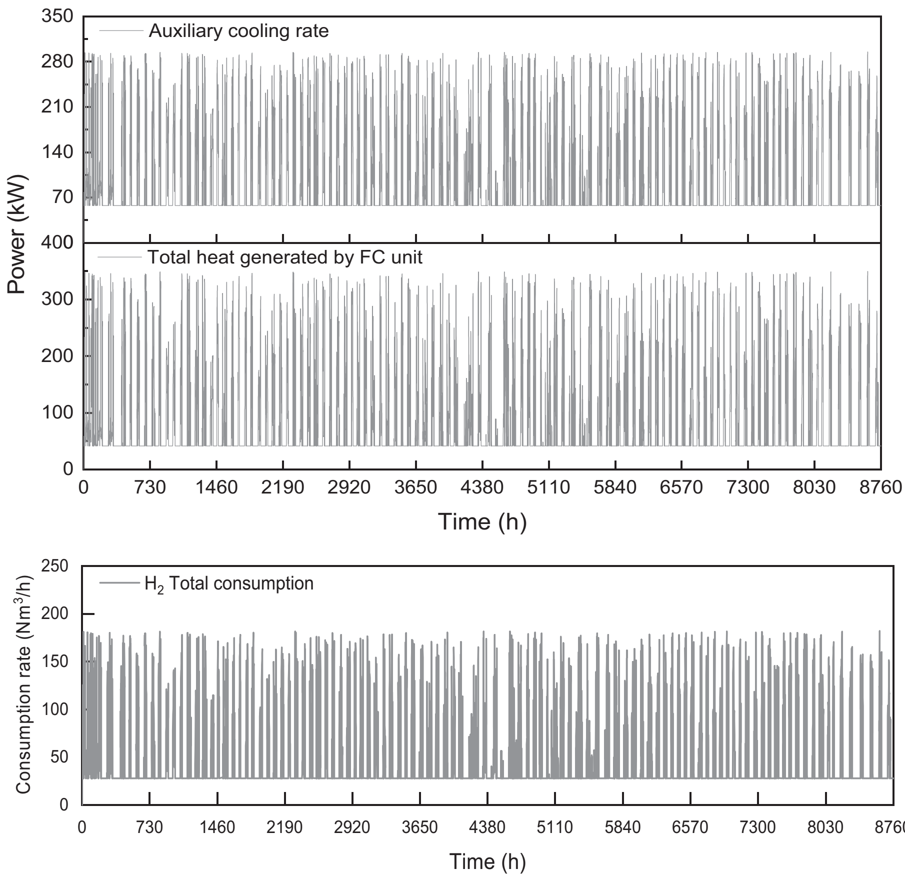

Fig. 9. Core outputs of the FC simulation result throughout the year.

energy. Finally, the Combined Dispatch strategy achieved an RF of 35.6 %, implying that 35.6 % of the energy produced in that mode was derived from renewable sources.

These comparative assessments shed light on the varying degrees of renewable energy penetration achieved in different studies. The proposed HRES in the current study demonstrates an RF of 44.57 %, indicating a relatively high utilization of renewable energy sources to meet the load requirements. This signifies the effectiveness and viability of the HRES in maximizing the use of renewable energy while minimizing reliance on conventional energy sources, thereby contributing to a more sustainable and environmentally friendly power generation system.

To provide a better insight into system performance, Fig. 12 depicts the amount of energy produced monthly by each constituent of the hybrid system, namely the WECS, DEGS, and AFC. It is noticeable that during the summer season, the diesel generators are seldom activated, especially during the period between April and September, which marks the time of greatest wind availability in the case study city, Dakhla (Morocco). Furthermore, the fuel cell is managed to operate only when necessary and therefore no excess energy is generated, in contrast to the wind turbine, which works all year round. In fact, the wind turbine output production is variable, due to the natural variability of the wind in Dakhla city, despite it being considered one of the windiest regions in Morocco. The capacity factor is hence mostly manifested by the availability of wind especially in hot seasons as seen in the results which may strengthen the fuel cell and electrolyzer operations to produce further hydrogen amounts in summer months.

The depicted variations in hydrogen volume and pressure in the storage tank provide valuable insights into the energy storage dynamics of the system Fig. 13. One notable observation is the peak in hydrogen volume during the summer season. This indicates that the excess energy generated by the system during this period is efficiently stored in the form of hydrogen. By storing the surplus energy, the system effectively captures and utilizes renewable energy during peak generation periods, ensuring its availability for later use when energy demand is higher or renewable energy generation is limited.

Additionally, the fluctuations in hydrogen pressure within the storage tank are worth considering. The significant drop in pressure during the winter season indicates that the stored hydrogen is being utilized to power the electrolysis system, which plays a crucial role in meeting energy demands during the colder months. This strategy enables the system to maintain a steady supply of energy even when renewable energy generation is reduced due to environmental factors.

By optimizing the storage and utilization of hydrogen, the hybrid system demonstrates its ability to effectively balance energy supply and demand throughout the year. The utilization of excess energy during peak generation periods and the subsequent use of stored hydrogen during periods of lower energy generation ensures a reliable and continuous energy supply. This dynamic approach enhances the system's resilience to seasonal variations and contributes to the overall stability and efficiency of the energy generation and storage process.

Fig. 14 presents the hourly variations in output power from the alkaline fuel cell (AFC) and negative input power (-pElectrolyzer) of the electrolyzer throughout the year. The plots show that the profiles of the fuel cell and the electrolyzer are almost inversely proportional. It is important to note that the rated output of the fuel cell and the electrolyzer is set at 300 kW and 500 kW, respectively. The system is designed to activate the electrolyzer only when there is a surplus of power (Wind-Load). When the surplus power in the microgrid falls below a critical level, the system turns off the electrolyzer and activates the diesel engine to meet the energy demand.

Based on the previous Fig. 12, a closer examination of the data presented in Fig. 14 reveals that the electrolyzer often had to operate at

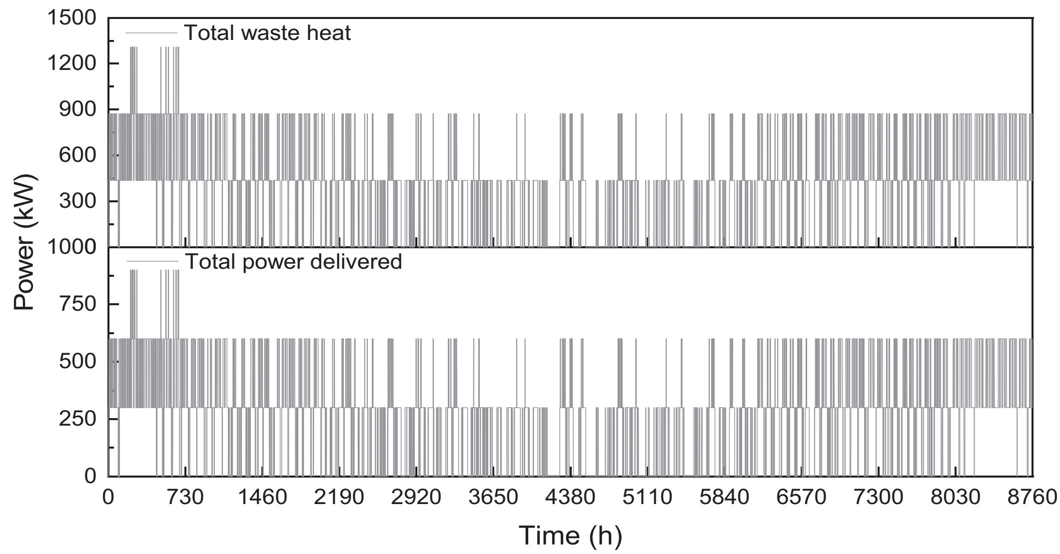

Fig. 10. Core outputs of DEGS simulation results throughout the year.

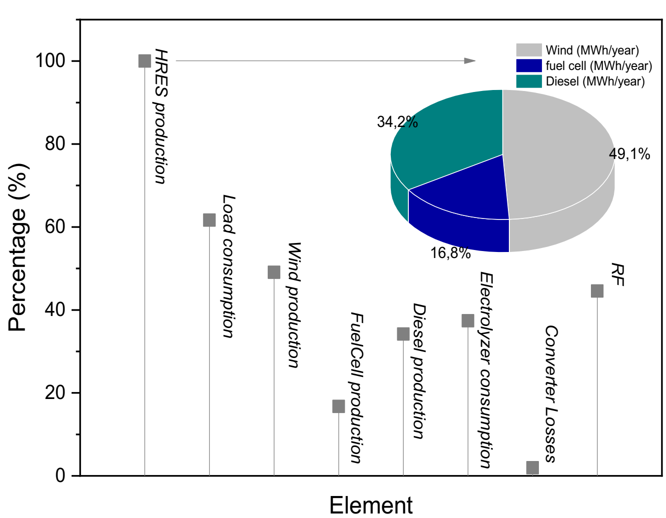

Fig. 11. The annual energy produced, consumed and lost with the RF of the optimal HRES.

reduced efficiency during the winter months due to the limited availability of wind, resulting in a lower surplus of energy to power the electrolysis process. As a result, the volume and pressure levels of hydrogen stored in the tank during the winter season were lower. However, the system is designed to address this by maximizing hydrogen production during the summer season when wind availability is high, and energy consumption is also high. The fuel cell and electrolyzer are programmed to operate inversely, with the electrolyzer only activated when there is excess energy available, and the fuel cell providing power when there is a shortage. The system is designed to ensure a dependable and sustainable power supply while reducing the use of fossil fuel-powered generators.

The detection of the stabilization of the hybrid system and ensuring sufficient hydrogen tank coverage for charging requirements is achieved by relying on the main state of charge (SOC) indication, which corresponds to the hydrogen density during the charging process up to the total hydrogen storage density (as shown in Fig. 15). The SOC of the hybrid system during the hot months is found to be slightly higher than that of the colder months, with approximately 0.52 in the hot months (June-July-August) compared to 0.18 in the colder months (November-January-February-March). The initial SOC value of the hybrid system is 0.50, which explains why the SOC was 50 % at the beginning of the year, as shown in Fig. 15. However, the monthly SOC evolution throughout the year tends to follow the trend of renewable energy supply, which increases in the summer due to the significance of wind during this period. Notably, the system does not stop charging the hydrogen tank until it reaches the specified setpoint when a specific setpoint is chosen. The CCM strategy applies this chosen dispatch strategy only when dispatchable components, such as the generator and fuel cell, are simultaneously operating during a time step.

Fig. 16 illustrates the pattern of variation for the required number of diesel generators (NDEGS) throughout the year. A closer examination of the plot reveals that there was low generator engagement, in months with high renewable energy (wind) that is responsible for power generation, and thus fewer diesel generators in operation. Conversely, higher generator engagement occurred during the months when renewable energy availability was lower. Hence resulting in a higher

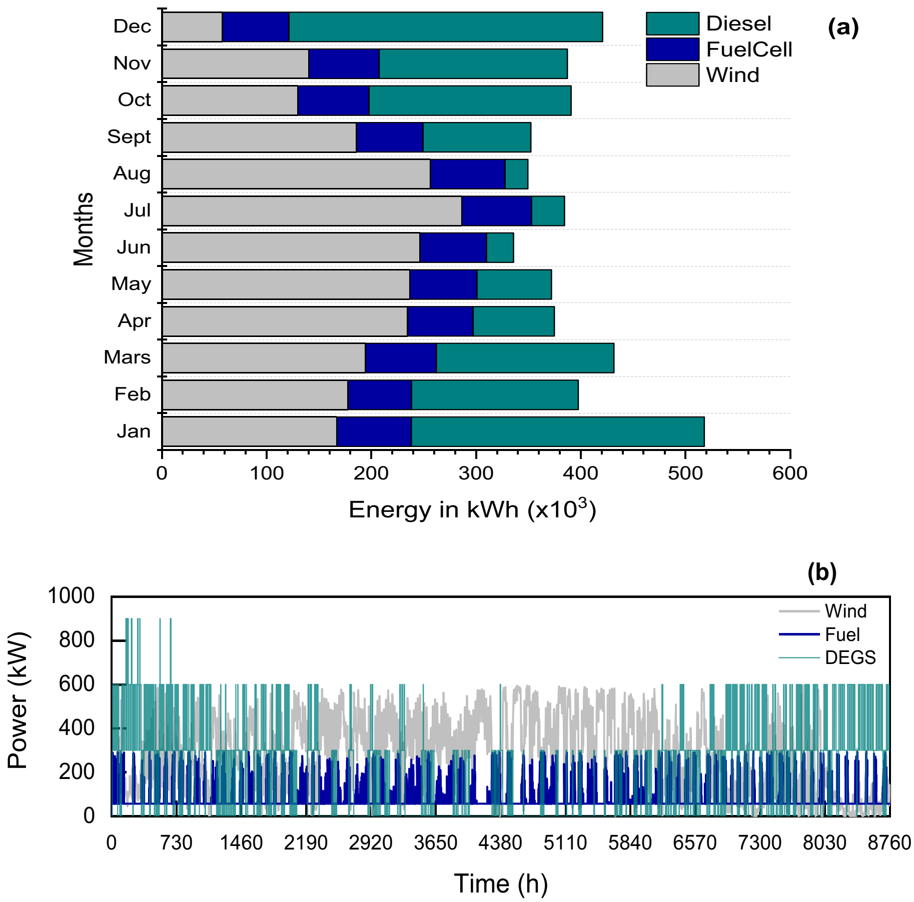

Fig. 12. Energy production by each element of the Hybrid system on a monthly (a) and hourly basis (b).

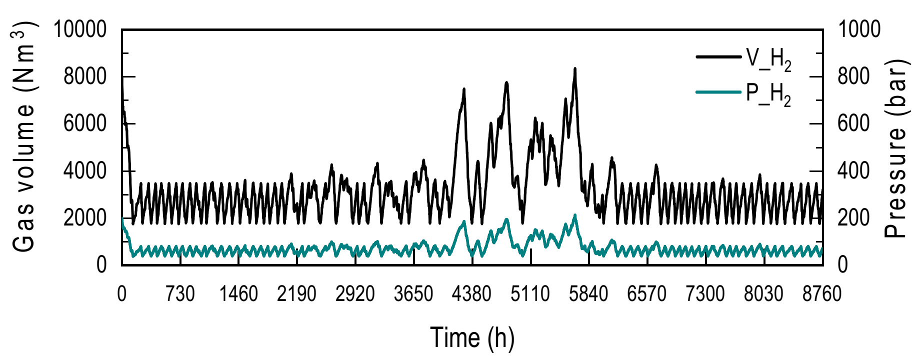

Fig. 13. Level of volume and pressure per hour of hydrogen storage of the optimal hybrid system over the year.

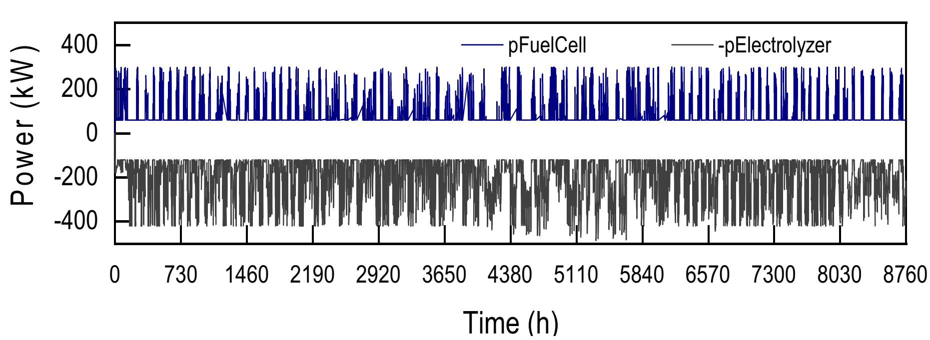

Fig. 14. The power produced per hour by the Alkaline Fuel Cell (pFuelcell) and the Electrolyzer's negative input power (-pElyctrolyzer) of the optimal hybrid system during the year.

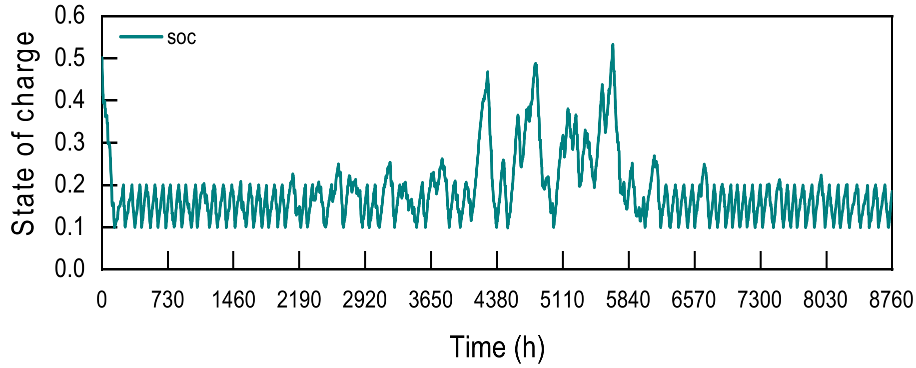

Fig. 15. State of charge variation of the hydrogen tank during the year.

contribution from diesel generators. This is seen especially in the cold winter months (November–January-February-march) wherein the number of generators needed is in the range of 1 to 3, unlike the warm seasons when renewable resource availability, such as wind in the current study is higher, the number of generators does not exceed 2 units. This process was a result of control adopted to cover the demand in times of shrinking renewable energy sources (especially in the winter season). The controller was set to run diesel generators to catch up with the reduction in wind availability, increasing the number of diesel generators needed (from 1 to 5 units) as backup power generation. Overall, the system was designed to optimize the use of renewable energy while ensuring a reliable and sustainable power supply.

After considering the factors discussed above, it is worth noting that hydrogen systems powered by wind (WECS) and paired with DEGS are one of the most attractive options for stand-alone power supply. Their modular nature allows for a versatile system that is advantageous for stand-alone power system setups. However, to determine the best system configuration, economic and environmental factors must be considered in the study of wind-based hydrogen systems. A thorough investigation of these factors will ensure that the system is not only cost-effective but also environmentally sustainable. Therefore, it is crucial to conduct a comprehensive evaluation of the economic and environmental impact of such systems before implementation.

5. Economical investigation

5.1. Technical and economic analysis

The economic analysis of a hybrid renewable energy system is an essential aspect to determine its financial and environmental viability. In this section, the performance of the proposed HRES partly met by DEGS as backup is analyzed with multiple evaluation criteria, which include power generation performance, losses energy, NPVCost, NPVProduction, LCOE, RF, and environmental impact. Therefore, in the optimization sizing and economic process all these parameters are taken into account involving the Renewable Fraction.

To perform the economic evaluation, several different hypotheses were made:

- The technical and economical details for each element of the HRES are presented in Table 4.
- The total investment capital, including all costs and the annual production energy of each element, is also provided (Table 5).
- The lifetime of the studied turbine is taken to be $n = 20$ years.
- The project's discount rate (r) has been set at 9.3 %.
- O&M Growth Rate fixed to 2 %

An Excel calculation tool was developed to estimate the net present value of the total cost and total electricity production, namely NPVCost and NPVProduction respectively, generated by the hybrid system over its lifespan. This tool was then used to compute the discounted cost of energy LCOE and the impact of carbon emissions. The inputs to the tool and the computation during the lifespan of the system were based on the technical and financial considerations of the system elements, as described in Table 4. Table 5 outlines the estimated costs and energy efficiencies for each element of the optimized hybrid system based on the available data. These cost and energy output values highlight the financial aspects and performance of the hybrid system components, providing a comprehensive understanding of their contributions to the overall system.

The replacement cost of each energy subsystem is assessed over its lifespan. Furthermore, the financial analysis takes into account the costs of operating and maintaining each element of the hybrid system. It is interesting to note that the operating cost of the diesel generator also includes the cost of natural gas over the lifespan of the generator.

The system cost estimations by component are shown in Fig. 17, including capital cost, replacement cost, operation and maintenance costs, natural gas, and salvage value. It is noteworthy that the operation

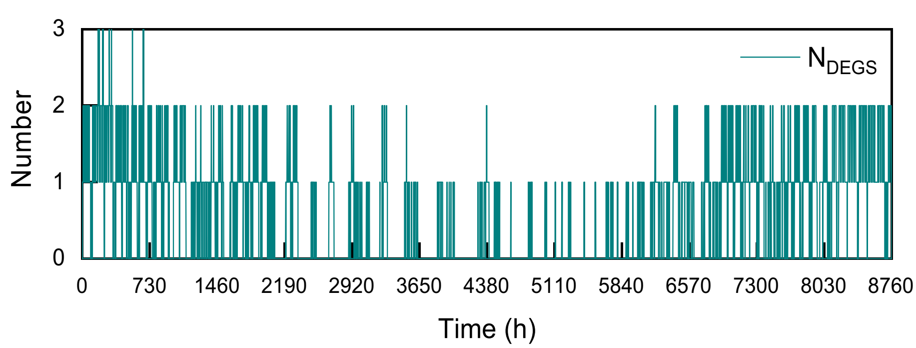

Fig. 16. Illustration of the necessary number of diesel (NDEGS) for the hybrid system during the year.

Table 4
Details on the technical and financial aspects of the hybrid system components.

	Maximum power output	Capital cost	Replacement	Operation and maintenance costs	Fuel cost	Lifespan	Capacity
Wind turbine	572.26 kW	\$1300/kW	\$1300/kW	\$26.7/kW	–	20 years	600 kW
Fuel cell	299.729 kW	\$600/kW	\$600/kW	\$0.01/h/kW = \$3.65/kW/year	–	10 years	300 kW
Hydrogen storage	750.5 (kg)	\$100/m^3	\$100/m^3	\$0.5/m^3	–	20 years	400 bar (50 m^3)
Electrolyzer	500 kW	\$151/kW	\$151/kW	\$8/year/kW	–	20 years	500 kW
Diesel generator (5 units)	1500 kW	\$230/kW	\$210/kW	\$0.007/kWh	\$0.1/m^3	15,000 h	300*5 = 1500 kW
Converter AC/DC	472.579 kW	\$550/kW	\$450/kW	\$5/kW/year	–	15 years	650 kW
Converter DC/AC	295.793 kW	\$550/kW	\$450/kW	\$5/kW/year	–	15 years	350 kW

Table 5
Overview of costs and energy outputs for all elements of the hybrid system.

Components	Cost of the initial investment	Costs of operations and maintenance	Fuel costs	Electricity production	Replacement
Wind turbine	\$743,938	\$15,279.34	–	2,313,504 kWh	\$0
Fuel cell	\$179,837.4	\$1094	–	812,398.9 kWh	\$179,837.4
Electrolyzer	\$75,500	\$4000	–	1,763,059 kWh	\$0
Hydrogen tank	\$4916	\$24.58	–	–	\$0
Natural gas (*5 DEGS*)	\$345,000	\$34,128	\$195	1,611,487.5 kWh (639.83 m^3 of natural gas used as a fuel)	\$0
Converters	259,918.45 + 162,686.15 = \$325,372.3	2362.895 + 1478.965 = \$3841.86	–	1,763,132 + 7,906,307 = 2,553,769 kWh	\$212,660.5 + \$133,106.85 = \$266,213.7
HRES	\$1,674,563.7	\$57,484	\$195	4,714,559 kWh	\$446,050.4

and replacement costs of the diesel generator units are higher than the capital cost. In addition, concerning other components, the cost of the wind turbine is more expensive, but this fact is expected to change in a short period of time, as the cost of renewable energy tends to decrease from year to year. This appears to be in line with the sort of cost reported in Fig. 18, showing that wind turbines are the most expensive component with a capital cost of \$743,938. However, the positive sign of the payback value (salvage value) suggests that there will be a return at the conclusion of the project's useful life, which is estimated to be \$356,772.85. This indicates that the system is financially feasible and can generate positive returns over its useful life. Overall, these findings emphasize the importance of conducting a thorough economic analysis that considers all relevant costs and factors in order to determine the financial viability of a hybrid renewable energy system.

5.2. Carbon emissions and levelized cost of energy impact

LCOE is an important parameter to consider when determining whether a company will build a project, as it must be cost-effective. This parameter can integrate fixed and variable costs into a single evaluation,

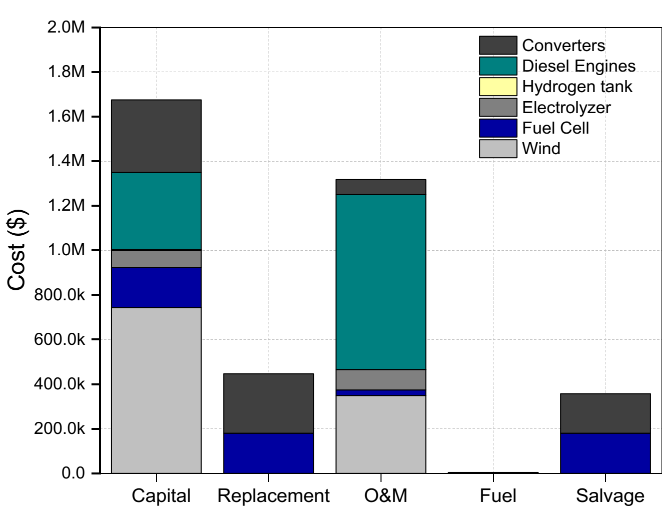

Fig. 17. An overview of the financial cost of the hybrid system's components.

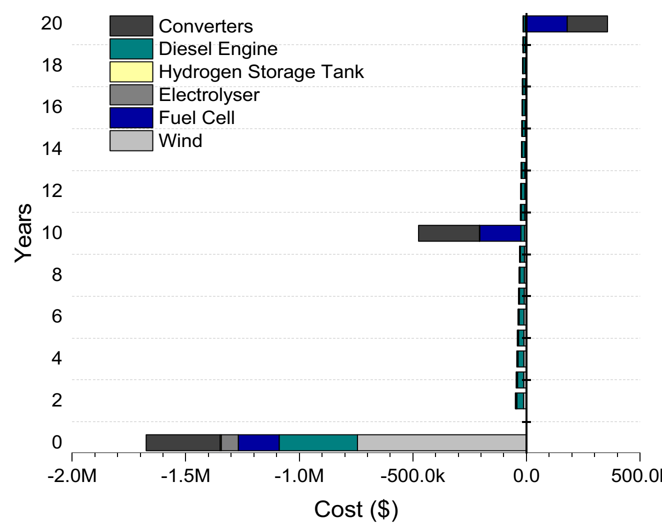

Fig. 18. Annual cash flow for each element during the lifespan of the hybrid system.

making it a useful tool to simplify the analysis. For LCOE calculation, a company must first identify critical factors, such as the system's lifespan, electricity production, and the input costs previously determined.

The Levelized Cost of Energy (LCOE) formula was incepted to economically examine the components of the proposed hybrid system by dividing the NPV_{Cost} (net present value of total cost) of power generated by the $NPV_{production}$ (net present value of total production) of the hybrid system over its lifespan.

$$LCOE = \frac{NPV_{Cost}}{NPV_{Production}} \tag{19}$$

where:

$$NPV_{Cost} = \sum_{t=1}^{n} \frac{I_t + O\&M_t + F_t}{(1+r)^t} \tag{20}$$

$$NPV_{Production} = \sum_{t=1}^{n} \frac{(E_{production})_t}{(1+r)^t} \tag{21}$$

I_t symbolizes the initial cost of the investment-related expenses, $O\&M_t$ represents the operation and maintenance expenses, it is to mention that during the lifespan of the hybrid system, the O&M expenses face an annual increase at a rate of 2 %. While F_t indicates the fuel costs applicable to diesel engines during the system's lifetime. Besides, the remaining two important coefficients to consider in the equation are the project discount rate (r), which is set at 9.3 %, and (n), which alludes to the hybrid system's life, which is considered to be 20 years.

All these variables are already determined and will be combined using the previously cited Eqs. (19), (20), and (21) to calculate the NPV_{cost}, $NPV_{production}$, and LCOE. Based on the calculations, the NPV_{Cost} and $NPV_{production}$ of the whole hybrid system were found to be respectively $2,650,843 and 37,819,172 kWh, resulting in an LCOE of the Hybrid System of $0.0701/kWh, as seen in Table 6. For more insight, Fig. 19 displays in detail the total outputs' net present value ($NPV_{production}$) for a project life of 20 years. The $NPV_{production}$ shows an exponential drop for all components, as anticipated over the lifespan of the hybrid system. Furthermore, if the drop in hybrid component prices continues, the proposed off-grid system will be able to compete with the grid electricity offered by the Moroccan government in the near future. Additionally, off-grid systems have the added benefit of contributing to the reduction of carbon dioxide emissions, which is particularly important for countries like Morocco that heavily depend on fossil fuel imports. However, it is important to note that the carbon footprint of off-grid systems is not zero, as they require energy and materials for manufacturing, transportation, and maintenance.

Meanwhile, the reduction in carbon dioxide emissions per year of a lifetime is an important techno-economic measure that also assesses the performance and environmental merits of the hybrid system. The estimation of the CO_2 emissions Factor in this study was based on IEA/IRENA Global renewable energy policies and database [42]. The CO_2 emissions are intended by taking the annual amounts of energy production during the lifespan of the hybrid system and then multiplying it by the emission factor (EF), this later is assumed in Morocco to be EF = 0.748 tCO_2/MWh. The proposed hybrid renewable system in this study has the potential to reduce CO_2 emissions by 70,529.80 tons during its 20-year life cycle of operation, with Total Carbon Abatement Costs (dollars per ton of carbon dioxide avoided) of about $37.5847 (Table 6).

6. Discussion and future perspectives

The study presented in this paper demonstrates the feasibility of using a hybrid renewable energy system (HRES) to provide reliable power to remote communities. The HRES was optimized using a power dispatch management strategy (PDMS) that involved load-following mode (LFM) and cycle charging mode (CCM) approaches. The PDMS approach in the current study was able to balance power generation and load demand, resulting in a system that met the power needs of the community with a high degree of reliability. The NPVcost and NPVproduction of the HRES were found to be $2,650,843 and 37,819,172 kWh, respectively, resulting in an LCOE of $0.0701/kWh. These results indicate that the proposed HRES can be a cost-effective alternative to grid electricity in remote off-grid areas. The renewable

Table 6
Financial results and emission reductions of CO_2 of the hybrid system.

NPV cost ($)	NPV production (kWh)	Levelized cost of energy ($/kWh)	Emission reductions over 20 years (t CO_2)	Total carbon abatement costs per t/CO_2e ($)
$2,650,843	37,819,172 kWh	$0.0701/kWh	70,529.8026 t CO_2	$37.5847

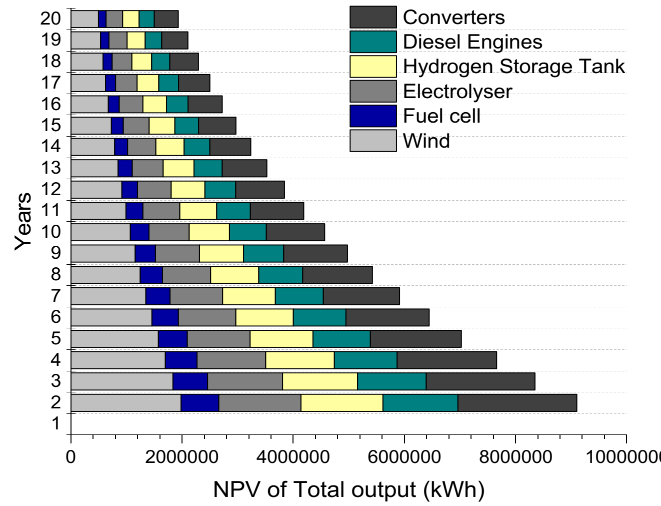

Fig. 19. NPV production concerning each component during the lifespan of the hybrid system.

fraction of 44.57 % also shows that a significant portion of the power needs of the community can be met using renewable sources. It is worth noting that the specific configuration presented in this study is tailored to the specific needs of the community in the case study. The optimal configuration may vary depending on the location, climate, and energy demand of different communities. However, the PDMS approach used in this study provides a framework that can be adapted and applied to other locations and scenarios to optimize HRES configurations.

In order to ensure the validity of the results presented in this study, a rigorous validation process was conducted (see Table 7). The numerical solution was validated by comparing it with the results obtained from a similar hybrid renewable energy system (HRES) deployed in the Sidi Daoud village in Tunisia, as reported in the study by Oueslati [30].

To perform the validation, the TRNSYS code was adjusted to match the optimal configuration derived from [30]. This configuration involved a combination of wind turbines, PV arrays, diesel engines, fuel cells, electrolyzers, hydrogen tanks, and converters. The load power demand and technical characteristics of each component were carefully calibrated within the TRNSYS code to ensure accurate representation.

The simulation results obtained from the TRNSYS code for the annual electricity outputs of each component were then compared with the corresponding results reported by [30]. Importantly, a high level of agreement was observed between the two sets of results, with a relative error of less than 1 %.

These successful validations provide strong evidence for the validity and accuracy of the numerical code and methodology employed in this study. The confidence level in the reliability of the results is significantly bolstered by these validation efforts, ensuring that the findings presented herein can be considered robust and trustworthy.

The proposed HRES system in the current study is expected to provide valuable technical and financial information for future applications of HRES in the Moroccan local environment, especially in regions with abundant renewable energy resources. The region under study has shown significant wind deposits, and it is also set to have wind projects in the future [43].

The obtained results demonstrate the feasibility and effectiveness of the hybrid renewable energy system (HRES) in meeting the energy demands of the targeted settlement. The system incorporates a wind turbine, fuel cell, electrolyzer, and diesel generators, working in harmony to ensure a reliable and sustainable power supply.

One notable aspect of the results is the efficient utilization of wind energy. When wind energy generation exceeds the demand, surplus energy is utilized for hydrogen production in the electrolyzer. This feature allows for energy storage in the form of hydrogen, which can be

Table 7
Comparison of annual energy outputs relative to all components of the optimized HRES, computed by the present code with those obtained by [30].

	PV panel	Wind turbine	Fuel cell	Electrolyzer	*DEGS*	Converters	HRES
Oueslati [30]	2,142,586 kWh	2,304,950 kWh	853,722 kWh	2,174,900 kWh	4,875,540 kWh	128,000 kWh	10,200,000 kWh
Present study	2,143,000 kWh	2,300,000 kWh	853,00 kWh	2,175,000 kWh	4,888,000 kWh	129,000 kWh	10,200,013 kWh
Relative error (%)	0.02 %	0.2 %	0.085 %	0.005 %	0.256 %	0.79 %	0.00013 %

later converted back to electricity using the fuel cell when wind power alone is insufficient to meet the load requirements. This dynamic energy management strategy ensures optimal utilization of renewable energy resources and minimizes reliance on backup diesel generators, reducing both costs and environmental impacts.

Regarding the economic analysis, the capital costs of the various system components have been evaluated. The wind turbine is found to have the highest capital cost, which is consistent with industry trends. However, it is worth noting that the cost of renewable energy technologies, including wind turbines, has been steadily decreasing over time, making them more economically viable. Therefore, while the initial investment for the wind turbine may be higher, it is expected to become more cost-effective in the long run.

The findings also indicate that the hybrid system is financially feasible, with positive returns expected over the system's useful life. The positive payback value, reflected in the salvage value of $356,772.85, suggests that there will be a return on investment at the conclusion of the project. This aspect is significant for stakeholders and decision-makers, as it highlights the potential economic benefits of adopting hybrid renewable energy systems.

Furthermore, the results emphasize the importance of conducting a thorough economic analysis that considers not only the initial capital costs but also the operation and maintenance costs, fuel costs, and potential revenue streams. This comprehensive approach ensures a more accurate assessment of the system's financial viability and enables informed decision-making.

Moreover, the study's findings can help inform future decisions regarding the deployment of HRES systems and the production of green hydrogen, making it a significant step in the region's transition to hybrid renewable energy sources. Especially that hydrogen is recognized globally as a crucial element in sustainable energy systems [44,45], with its potential to reduce carbon dioxide emissions and its versatility in different energy sectors. The study revealed that the proposed hybrid renewable energy system has the potential to significantly reduce CO_2 emissions by 70,529.80 metric tons over a 20-year period, with an estimated carbon reduction cost of $37.5847 per metric ton. This translates to a total cost savings of approximately $2,656,754.51, indicating a substantial positive impact on the environment.

Given that Morocco does not have a specific nationwide carbon pricing policy or carbon tax implemented yet [46], the study makes use of an estimation based on the assumption that the environmental cost of each kilogram of CO_2 emissions is equivalent to 0.125 dollars. This estimation draws inspiration from a similar approach used in Sweden [47]. However, it is important to note that numerous experts argue for a significantly higher true cost. Estimates vary widely, ranging from $100 to $200 per ton [50]. By employing this estimation, it becomes evident that the environmental advantages of implementing the hybrid renewable energy system go beyond mere cost savings when compared to a traditional system. The implementation of the proposed hybrid system is projected to generate an equivalent green space that would be comparable to adding approximately 5256.16 acres (assuming 1 acre = 4046.86 square meters), approximately 2125 ha of new vegetation and plants, as illustrated in Fig. 20. Therefore, the implementation of this system can not only provide a sustainable and reliable source of electricity but also contribute to mitigating the effects of climate change.

The study's findings have significant implications, showcasing the potential of HRES systems to provide reliable power to remote communities and reduce carbon dioxide emissions, which is essential for the transition to a low-carbon economy. This is particularly relevant for Morocco, being one of the select few countries with substantial capacity for green hydrogen production, making it capable of meeting 4 % of global demand by 2030, according to the World Energy Council [43]. Therefore, the findings of this study hold particular importance.

The proposed PDMS approach offers optimization for HRES systems, enhancing their efficiency and cost-effectiveness. Additionally, this study contributes to the ongoing efforts in the development of green hydrogen production and the deployment of HRES systems, both of which are crucial components of sustainable energy systems. Hence, the proposed solutions have the potential to be a formidable competitor to grid connections. Especially as unit prices for renewable energy technologies continue to decrease. This presents exciting opportunities for off-grid electricity generation, particularly in African rural areas that have abundant renewable energy resources. By implementing these solutions, the challenges surrounding electricity access in these regions can be addressed effectively. The availability of renewable energy sources, such as solar and wind power, in African rural zones provides untapped potential for the deployment of off-grid hybrid renewable energy systems. These systems can harness abundant renewable resources to generate clean and reliable electricity, enabling communities to overcome the limitations of traditional grid infrastructure. Consequently, the study's findings can inform future decision-making processes concerning the implementation of HRES systems and the production of green hydrogen, representing a significant step toward the transition to hybrid renewable energy sources.

7. Conclusion

Upon several bases set, this paper highlights a new power dispatch management strategy (PDMS) that combines load following mode (LFM) and cycle charging mode (CCM) approaches to analyze a hybrid renewable energy system using hydrogen as a fuel. The study focused on meeting the energy load demands of a settlement in Dakhla City.

To study the dynamic behavior of the entire system, a simulation program using TRNSYS software was carried out, giving an overview of the performance behavior of each component of the HERS while ensuring the stability of the system at all times. The results underscored the dependence of the system configuration's attractiveness on factors such as management strategy, minigrid controller, wind availability, hydrogen storage, and auxiliary backup energy.

Furthermore, a comprehensive technical-economic analysis was conducted to assess various aspects of system component costs. This analysis encompassed capital, replacement, operation, and maintenance (O&M), and fuel costs, as well as the payback value. Financial metrics such as yearly cash flow, net present value (NPV) of total outputs and production, levelized cost of electricity (LCOE), and CO_2 emission reduction with its Total Abatement Costs were also taken into consideration.

The main findings of the numerical modeling of the HRES are as follows:

i. The proposed HRES system has shown a significant potential to meet the load requirement of the settlement, with significant penetration of renewable energy by exhibiting a good renewable fraction (44.57 %).
ii. The simulation results show that the hybrid system with a 600 kW wind turbine, five 300 kW diesel generators, a 300 kW fuel cell,

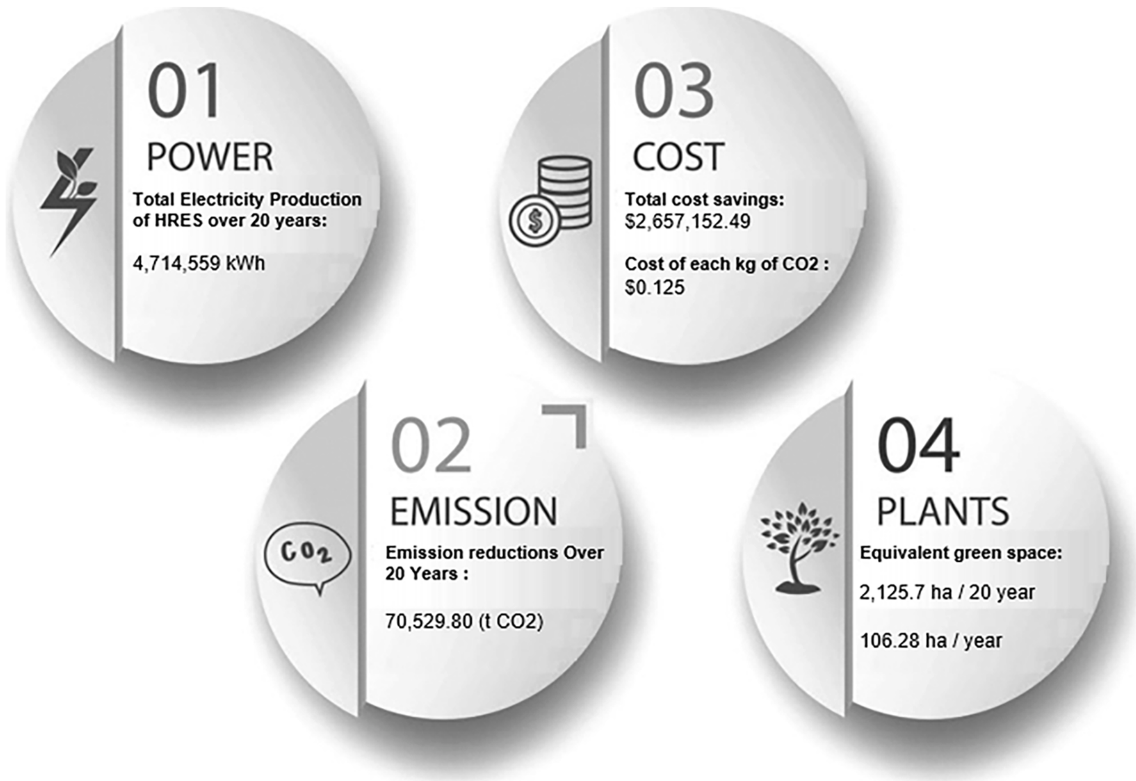

Fig. 20. Illustrates the techno-economic-environmental advantages of the proposed HRES for the settlement community.

and a 500 kW electrolyzer is the most cost-effective hybrid system, with the net present value of total cost (NPV_{Cost}) and the net present value of total production ($NPV_{production}$), respectively, were $2,650,843 and 37,819,172 kWh, resulting in an LCOE of $0.0701/kWh for the hybrid system.

iii. The finding also reveals the ability to cut CO_2 emissions by 70,529.80 tons for a 20-year working life cycle, with overall carbon reduction costs per t/CO_2e of about $37.5847. The presented HRES can then be more competitive by taking into account the continuing decrease in the cost of renewable technologies, which might lead to even more significant reductions in the cost of the suggested hybrid system in the not-to-distant term.

iv. The implementation of the proposed hybrid system is projected to generate an equivalent green space that would be comparable to adding approximately 5256.16 acres (approximately 2125 ha) of new vegetation and plants, based on the assumption that 1 acre is equivalent to 4046.86 square meters.

According to the simulation results, the implementation of this system offers several significant benefits. Not only does it provide a sustainable and dependable source of electricity, but it also plays a crucial role in mitigating the adverse effects of climate change. The analysis revealed that wind hydrogen production in the Dakhla region was highly satisfactory, indicating that it is a suitable location for harnessing autonomous hydrogen derived from wind energy. This presents a tremendous opportunity to electrify remote communities with minimal energy production costs, while simultaneously reducing their environmental footprint and addressing the energy crisis. Ultimately, this system contributes to a sustainable future by promoting cleaner energy solutions and improving overall energy sustainability.

CRediT authorship contribution statement

Sara El Hassani: Conceptualization, Methodology, Simulation, Formal analysis, Investigation, Validation. **Fakher Ouesselati**: Conceptualization, Formal analysis, Investigation, **Othmane Horma**: Writing -review & editing. **Domingo Santana**: review & editing, Supervision. **Mohammed Amine Moussaoui**: Investigation, Writing -review & editing, Supervision. **Ahmed Mezrhab**: Writing- review & editing, supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] A. Bemporad, M. Morari, Control of systems integrating logic, dynamics, and constraints, Automatica 35 (3) (1999) 407–427.
[2] M. Alonso, H. Amaris, C. Alvarez-Ortega, Integration of renewable energy sources in smart grids by means of evolutionary optimization algorithms, Expert Syst. Appl. 39 (5) (2012) 5513–5522.
[3] S.E. Hassani, H.A.L. Ouali, M.A. Moussaoui, A. Mezrhab, Techno-economic analysis of a hybrid CSP/PV plants and A. S. E. in the eastern region of Morocco, Vol. 57, No. 4, pp. 297–309, 2021, "Techno-economic analysis of a hybrid CSP/PV plants in the eastern region of Morocco", Appl. Solar Energy 57 (4) (2021) 297–309.
[4] S. El Hassani, O. Horma, O. Selhi, M.A. Moussaoui, A. Mezrhab, Modeling and evaluating concentrated solar power plant performance in Morocco: a parametric study, Int. J. Energy Clean Environ. 24 (2023).
[5] L. Bousselamti, M. Cherkaoui, Modelling and assessing the performance of hybrid PV-CSP plants in Morocco: a parametric study, Int. J. Photoenergy 2019 (2019).
[6] C. Parrado, A. Girard, F. Simon, E. Fuentealba, 2050 LCOE (levelized cost of energy) projection for a hybrid PV (photovoltaic)-CSP (concentrated solar power) plant in the Atacama Desert, Chile, Energy 94 (2016) 422–430.
[7] S. Barakat, A. Emam, M.M. Samy, Investigating grid-connected green power systems' energy storage solutions in the event of frequent blackouts, Energy Rep. J. 8 (November 2022) 5177–5191 (Elsevier).

[8] M.M. Samy, A. Emam, Elsayed Tag-Eldin, S. Barakat, Exploring energy storage methods for grid-connected clean power plants in case of repetitive outages, J. Energy Storage 54 (2022), 105307, https://doi.org/10.1016/j.est.2022.105307.
[9] M.B. Eteiba, S. Barakat, M.M. Samy, W.I. Wahba, Optimization of an off-grid PV/biomass hybrid system with different battery technologies, Sustain. Cities Soc. 40 (2018) 713–727, https://doi.org/10.1016/j.scs.2018.01.012.
[10] S. Tetteh, M.R. Yazdani, A. Santasalo-Aarnio, Cost-effective electro-thermal energy storage to balance small scale renewable energy systems, J. Energy Storage 41 (2021), 102829 (2021).
[11] Y. Jiao, D. Månsson, Greenhouse gas emissions from hybrid energy storage systems in future 100% renewable power systems–a Swedish case based on consequential life cycle assessment, J. Energy Storage 57 (2023), 106167.
[12] P. Marocco, R. Novo, A. Lanzini, G. Mattiazzo, M. Santarelli, Towards 100% renewable energy systems: the role of hydrogen and batteries, J. Energy Storage 57 (2023), 106306.
[13] J. Lokar, P. Virtič, The potential for integration of hydrogen for complete energy self-sufficiency in residential buildings with photovoltaic and battery storage systems, Int. J. Hydrog. Energy 45 (60) (2020) 34566–34578.
[14] B.D. Mert, F. Ekinci, T. Demirdelen, Effect of partial shading conditions on off-grid solar PV/hydrogen production in high solar energy index regions, Int. J. Hydrog. Energy 44 (51) (2019) 27713–27725.
[15] Q. Hassan, I.S. Abdulrahman, H.M. Salman, O.T. Olapade, M. Jaszczur, Techno-economic assessment of green hydrogen production by an off-grid photovoltaic energy system, Energies 16 (2) (2023) 744.
[16] M.S. Saleem, et al., Design and optimization of hybrid solar-hydrogen generation system using TRNSYS, Int. J. Hydrog. Energy 45 (32) (2020) 15814–15830.
[17] C. Ghenai, T. Salameh, A. Merabet, Technico-economic analysis of off grid solar PV/fuel cell energy system for residential community in desert region, Int. J. Hydrog. Energy 45 (20) (2020) 11460–11470.
[18] H.M. Al Ghaithi, G.P. Fotis, V. Vita, Techno-economic assessment of hybrid energy off-grid system—a case study for Masirah island in Oman, Int. J. Power Energy Res. 1 (2) (2017) 103–116.
[19] M. Qolipour, A. Mostafaeipour, O.M. Tousi, Techno-economic feasibility of a photovoltaic-wind power plant construction for electric and hydrogen production: a case study, Renew. Sust. Energ. Rev. 78 (2017) 113–123.
[20] H. Hassani, F. Zaouche, D. Rekioua, S. Belaid, T. Rekioua, S. Bacha, Feasibility of a standalone photovoltaic/battery system with hydrogen production, J. Energy Storage 31 (2020), 101644.
[21] R. Loisel, L. Baranger, N. Chemouri, S. Spinu, S. Pardo, Economic evaluation of hybrid off-shore wind power and hydrogen storage system, Int. J. Hydrog. Energy 40 (21) (2015) 6727–6739.
[22] M. Gökçek, C. Kale, Techno-economical evaluation of a hydrogen refuelling station powered by Wind-PV hybrid power system: a case study for İzmir-Çeşme, Int. J. Hydrog. Energy 43 (23) (2018) 10615–10625.
[23] S. Shaahid, et al., Techno-economic potential of retrofitting diesel power systems with hybrid wind-photovoltaic-diesel systems for off-grid electrification of remote villages of Saudi Arabia, Int. J. Green Energy 7 (6) (2010) 632–646.
[24] S. Silva, M. Severino, M. De Oliveira, A stand-alone hybrid photovoltaic, fuel cell and battery system: a case study of Tocantins, Brazil, Renew. Energy 57 (2013) 384–389.
[25] M. Rezaei, A. Mostafaeipour, M. Qolipour, H.-R. Arabnia, Hydrogen production using wind energy from sea water: a case study on southern and northern coasts of Iran, Energy Environ. 29 (3) (2018) 333–357.
[26] A.F. Tazy, M.M. Samy, S. Barakat, A techno-economic feasibility analysis of an autonomous hybrid renewable energy sources for university building at Saudi Arabia, J. Electr. Eng. Technol. 15 (2020) 2519–2527.
[27] Charafeddine Mokhtara, Belkhir Negrou, Noureddine Settou, Belkhir Settou, Mohamed Mahmoud Samy, Design optimization of off-grid hybrid renewable energy systems considering the effects of building energy performance and climate change: case study of Algeria energy, Energy Int. J. 219 (15) (March 2021), 119605 (Elsevier).
[28] S. Rehman, M.M. Alam, J.P. Meyer, L.M. Al-Hadhrami, Feasibility study of a wind-pv-diesel hybrid power system for a village, Renew. Energy 38 (1) (Feb 2012) 258–268, https://doi.org/10.1016/j.renene.2011.06.028.
[29] F. Dawood, G. Shafiullah, M. Anda, Stand-alone microgrid with 100% renewable energy: a case study with hybrid solar PV-battery-hydrogen, Sustainability 12 (5) (2020) 2047.
[30] F. Oueslati, Hybrid renewable system based on solar wind and fuel cell energies coupled with diesel engines for Tunisian climate: TRNSYS simulation and economic assessment, Int. J. Green Energy 18 (4) (2021) 402–423.
[31] M.M. Samy, Techno-economic analysis of hybrid renewable energy systems for electrification of rustic area in Egypt, Innov. Syst. Des. Eng. 8 (1) (2017) 42–54.
[32] M.M. Samy, N. Alshammari, J. Asumadu, Optimal economic analysis study for renewable energy systems to electrify remote region in Kingdom of Saudi Arabia, in: The Twentieth International Middle East Power Systems Conference (MEPCON), Cario University, Egypt, 2018, pp. 1040–1045 (December 18-20).
[33] Aykut Fatih Guven, M.M. Samy, Performance analysis of autonomous green energy system based on multi and hybrid metaheuristic optimization approaches, Energy Convers. Manag. 269 (2022), 116058, https://doi.org/10.1016/j.enconman.2022.116058.
[34] S. Hajiaghasi, A. Salemnia, M. Hamzeh, Hybrid energy storage system for microgrids applications: a review, J. Energy Storage 21 (2019) 543–570.
[35] A. Jani, H. Karimi, S. Jadid, Hybrid energy management for islanded networked microgrids considering battery energy storage and wasted energy, J. Energy Storage 40 (2021), 102700.
[36] M.M. Samy, Heba I. Elkhouly, S. Barakat, Multi-objective optimization of hybrid renewable energy system based on biomass and fuel cells, Int. J. Energy Res. 45 (6) (May 2021).
[37] M.M. Samy, S. Barakat, H.S. Ramadan, A flower pollination optimization algorithm for an off-grid PV-fuel cell hybrid renewable system, Int. J. Hydrog. Energy 44 (4) (2019) 2141–2152.
[38] Aykut Fatih Guven, Nuran Yorukeren, M.M. Samy, Design optimization of a stand-alone green energy system of university campus based on JAYA-harmony search and ant colony optimization algorithms approaches, Energy J. 253 (2022) (2022), 124089, https://doi.org/10.1016/j.energy.2022.124089 (Elsevier).
[39] M.M. Samy, Rabia Emhamed Almamlook, Heba I. Elkhouly, S. Barakat, Design decision-making and optimal design of green energy system based on statistical methods and artificial neural network approaches, Sustain. Cites Soc. J. 84 (September 2022), 104015, https://doi.org/10.1016/j.scs.2022.104015 (Elsevier).
[40] *TRNSYS, A transient simulation program (Version 18) manual.*
[41] M. Boussetta, R. El Bachtiri, M. Khanfara, K. El Hammoumi, Assessing the potential of hybrid PV–wind systems to cover public facilities loads under different Moroccan climate conditions, Sustain. Energy Technol. Assess. 22 (2017) 74–82.
[42] [Online] Available: https://www.irena.org/IRENADocuments/Statistical_Profiles/Africa/Morocco_Africa_RE_SP.pdf.
[43] [Online] Available: https://energynews.biz/dakhla-city-could-produce-green-hydrogen-at-the-lowest-cost-study-finds/.
[44] A. Nieto, V. Vita, T.I. Maris, Power quality improvement in power grids with the integration of energy storage systems, Int. J. Eng. Res. Technol. 5 (7) (2016) 438–443.
[45] S. Lazarou, V. Vita, M. Diamantaki, D. Karanikolou-Karra, G. Fragoyiannis, S. Makridis, L. Ekonomou, A simulated roadmap of hydrogen technology contribution to climate change mitigation based on Representative Concentration Pathways considerations, Energy Sci. Eng. 6 (3) (2018) 116–125.
[46] [Online] Available: https://read.oecd-ilibrary.org/taxation/pricing-greenhouse-gas-emissions_023747ff-en#page25.
[47] Sweden's Carbon Tax Government.se. https://www.government.se/government-policy/swedens-carbon-tax/swedens-carbon-tax/, 2022. Accessed 28th Jun 2022.
[48] A.B. Awan, M. Zubair, G.A.S. Sidhu, A.R. Bhatti, A.G. Abo-Khalil, Performance analysis of va ious hybrid renewable energy systems using battery, hydrogen, and pumped hydro-based storage units, Int. J. Energy Res. 43 (2019) 6296–6321, https://doi.org/10.1002/er.4343.
[49] A.S. Aziz, M.F.N. Tajuddin, M.R. Adzman, M.A.M. Ramli, S. Mekhilef, Energy management and optimization of a PV/diesel/battery hybrid energy system using a combined dispatch strategy, Sustainability 11 (2019) 683, https://doi.org/10.3390/su11030683.
[50] P.H. Howard, D. Sylvan, The Economic Climate: Establishing Consensus on the Economics of Climate Change, No. 330-2016-13703, 2015, pp. 1–77.
[51] S. El Hassani, M. Charai, M.A. Moussaoui, A. Mezrhab, Towards rural net-zero energy buildings through integration of photovoltaic systems within bio-based earth houses: Case study in Eastern Morocco, Solar Energy 259 (2023) 15–29.